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EDF-X API Guide

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Table of Contents

1	Document Structure	5
2	Introduction.....	7
3	Accessing the EDF-X API.....	9
3.1	Base URL	9
3.2	Response codes.....	9
3.3	Authentication	9
4	Searching For An Entity.....	11
4.1	Entity Search Endpoint.....	11
4.2	Batch Entity Search Endpoint	13
5	Retrieving PD Values.....	17
5.1	Request Structure.....	17
5.1.1	Enabling Further Detail.....	19
5.1.2	Using Model Parameters for Advanced Users	19
5.1.3	Using in Asynchronous Mode.....	19
5.1.4	Using processId	20
5.2	Response Structure	20
5.2.1	Response Sections.....	20
6	Uploading Company Information.....	23
6.1	Download the template.....	24
6.2	Upload the financial statements.....	24
7	Accessing Processes	25
7.1	Status.....	25
7.2	Files	26
8	Retrieving Limits for Trade Credit	27
9	Retrieving Peer Groups	29
9.1	IDs.....	29
9.2	Metrics.....	31
9.3	Percentile	32
9.4	Metadata	33
9.5	Recommended	34
10	Retrieving Early Warning Category and Trigger	35
10.1	Risk Category.....	35
10.2	Triggers.....	37
11	Retrieving Financial Information and Ratios	41
11.1	Statements.....	41
11.2	Ratios.....	43
11.3	Ratio Calculation	45

12	Frequently Asked Questions.....	49
12.1	Data.....	49
12.2	Scoring Process.....	52
12.3	Results	52
13	Appendix: Input Details.....	57
13.1	Universal Input Template - Minimal	57
13.2	Universal Input Template - Full.....	59
13.3	Banks Input Template.....	63
13.4	NDY - List of values	67
14	Appendix: Output Details	71
14.1	PD Analytics	71
14.1.1	Standard Outputs.....	72
14.1.2	Entity and Country-Specific Outputs	74
14.2	Credit Limits.....	79
15	Appendix: Models Inventory.....	81
16	Appendix: Peer Groups - Country to Region Mapping.....	83
17	Appendix: Supported Identifiers	93

1 Document Structure

This guide describes relevant information to use and understand the EDF-X API. It has the following structure:

- [Accessing the EDF-X API](#): Provides the location of the end points used and how to authenticate using a username and password
- [Entity Search](#): Explains how users can search for an entity
- [Retrieving PD values](#): Explains how users can access PD values
- [Uploading Company Information](#): Explain show users can upload their own data
- [Accessing Processes](#): Explains how to monitor and manage processes initiated by other endpoints
- [Retrieving Limits for Trade Credit](#): Explains how to access limits for trade credit lending
- [Retrieving Peer Group](#): Explains how to get information about entities' peer groups
- [Retrieving Risk Category and Triggers](#): Explains how to get the risk category of an entity
- [Retrieving Financial Information and Ratios](#): Explains how to get the financial statements and the related key ratios
- [Frequently Asked Questions](#): Answers to some frequently asked questions

The following appendices provide additional detail:

- [Input Fields](#): Provides details of the inputs required by the API
- [Appendix: Output Details](#): Describes the output produced
- [Model Inventory](#): Lists the models available
- [Country to Region](#): Mapping of each country to the corresponding region, this is relevant for peer group endpoints
- [Supported Identifiers](#): Lists the identifiers that can be used in the Search endpoints

2 Introduction

Moody's Analytics has a long, successful track record of producing industry-leading models and analytics from its rich proprietary dataset of company financials, defaults, and recovery. The Moody's Analytics PD is our best estimate of probability of default (PD), given the information available. Our PD values are based on years of experience and provide clients with an unparalleled combination of quality and reach. The EDF-X API provides easy access to these calculation methods and directly to PDs for the approximately 400 million (publicly listed and private) companies in Orbis globally.

The EDF-X API expects one of two types of input:

- If the company is one of the approximately 400 million we have details on, then all you must do is send an identifier and we send you our best estimate of credit risk.
- If the company is not – or you would prefer to use your own data – then the API acts as a credit risk calculator based on your input. The most appropriate model is selected based on the input provided and the corresponding PD is generated.

Based on the input provided, the EDF-X API provides the access to following outputs:

- A PD for any company. This is our best-estimate of the one-year probability of default given the information available.
- A PD term structure providing annualized, cumulative, and forward PD values out to ten years
- An implied rating indicating the Moody's Analytics rating with a consistent default profile.
- A confidence indicator explaining which methodology and data were used to derive the estimate. This helps the end user understand and assess their confidence in the estimate.
- Full details of the model calculation, including input variables and links to the relevant methodology document.
- Limits for trade credit corresponding to conservative, balanced, and aggressive assumptions.
- The peer groups a company belongs to as well as their main traits.
- The appropriate Early Warning category for the company, as well as PD thresholds that are considered as triggers in the calculation.

Additional end points help the user:

- Search for entity identifiers
- Access peer group metrics
- Access financial statements and key ratios

3 Accessing the EDF-X API

3.1 Base URL

EDF-X API end points are all relative to the base URL:

```
https://api.edfx.moodyanalytics.com/edfx/v1/
```

Combining a base URL with a specific endpoint enables the API to access a specific resource.

For example, the URL for the entities/pds endpoint is:

```
https://api.edfx.moodyanalytics.com/edfx/v1/entities/pds
```

3.2 Response codes

The following table provides the EDF-X endpoint response codes and their descriptions.

EDF-X Response Codes

Response Code	Description	Notes
200	OK	
201	Response created	Request parameter async must be true
202	Accepted - processing in progress	Request parameter async must be true
401	Unauthorized - Provided Token is Invalid	
4XX	Parameter Request Error - the request parameters are incorrect	
5XX	Server Error - A problem has been found with the system	Contact MA Support

3.3 Authentication

The EDF-X API uses the OAuth 2.0 protocol using the password flow authentication mechanism. Each API request to the EDF-X API requires a bearer token. To generate the token, you must make a request to the authorization token endpoint using the following details:

Note The response provides an `id_token`. The response is used in the following format `Bearer <id_token>` within the authorization header of the request.

- Method: POST
- URL: `https://sso.moodyanalytics.com/sso-api/v1/token`

Request Parameters

Parameter	Value	Data type	Description	Parameter type
username	Username	String	The provided username.	Part of the form field
password	Password	String	The provided password.	Part of the form field
grant_type	"password"	String	The OAuth password flow.	Part of the form field
scope	"openid"	String	The OAuth scope.	Part of the form field

The Authentication service has the following response codes.

HTTP Response

Response code	Description
200	OK - User Authenticated
401	Unauthorized - Provided credentials are not valid
4XX	Parameter Request Error - The request parameters are incorrect
5XX	Server Error - A problem has been found with the system

Example Request

```
curl --location --request POST 'https://sso.moodyanalytics.com/sso-api/v1/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'username=<Your username here>' \
--data-urlencode 'password=<Your password here>' \
--data-urlencode 'grant_type=password' \
--data-urlencode 'scope=openid'
```

Example Response

```
{
  "access_token": "96fbbb9c-ef70-3ab8-b847-1e92eaa1c8bd",
  "id_token": "eyJraWQiOiJNbUp etc.",
  "token_type": "Bearer"
}
```

4 Searching For An Entity

To retrieve pre-calculated PD values, you must provide company identifiers, either the entity's CreditEdge PID or BvD Identifier (BVDID). The Entity Identifier Search Tool can be used to find the BVDID. By providing the company name or one of the specified identifiers, you can retrieve a BVDID which can be used in the EDF-X API. Where more than one company matches the search, the search returns a list of BVDIDs.

Available Endpoints

End Point	Method	Description
/entity/v1/search	POST	Use this endpoint to search for one company.
/entity/v1/mapping	POST	Use this endpoint to search for a batch of companies.

4.1 Entity Search Endpoint

Endpoint: /entity/v1/search

For the URL, combine the base URL ([Accessing the EDF-X API](#)) with the endpoint.

Method: POST

Request Parameters

Name	Required?	Description	Preset Value	Example
query	Yes	A string containing a company identifier. The currently supported identifiers are listed in "Appendix: Supported Identifiers"; the main ones are: - Name (exact company name) - BVDID - Identifier from Orbis - ISIN - International Securities Identification Number - CUSIP - Committee on Uniform Securities Identification Procedures - LEI - Legal Entity Identifier - PID - CreditEdge Identifier Provide the value. The API will automatically determine which identifier was sent based on its form.	N/A	"Apple Inc"
limit	No	Specify the maximum number of results returned in single call.	20	40

Name	Required?	Description	Preset Value	Example
offset	No	Specify the index from which to start returning results. Use with the limit parameter to get result batches to process. For example, limit= 10, offset = 20, returns the results indexed from 20-29.	0	10

Sample JSON request

```
{
  "query": "Apple Inc"
}
```

Response

The API returns a list of identifiers for the companies that match the search. The information that you can expect to see varies depending on the data available for the entity, in general:

- identifiers related to the Orbis database or Moody's Analytics identifiers: Orbis ID, BVD ID, PID (for public firms)
- international or national identifiers: ISIN, CUSIP, LEI, tax number, ticker, etc.
- firmographic information: firm name, country, city, address, industry (according to different industry classifications), etc.
- peer groups: IDs of the peer groups the firm belongs to. These are pre-canned peer groups generated and owned by Moody's Analytics, ranked by relevance. Refer to [Retrieve Peer Groups](#) to discover more.

Sample JSON response

```
{
  "entities": [
    {
      "entityId": "US942404110",
      "pid": "037833",
      "identifierOrbis": "000411117",
      "identifierBvd": "US942404110",
      "primaryIndustryNACE": "2630",
      "primaryIndustryNAICS": "334220",
      "primaryIndustryNDY": "N21",
      "primaryIndustrySIC": "3663",
      "primaryIndustryNDYDescription": "ELECTRONIC EQUIPMENT",
      "type": "Corporate",
      "entityType": "Corporate",
      "internationalName": "APPLE, INC.",
      "contactAddressLine": "One Apple Park Way",
      "contactCountryCode": "USA",
      "contactPostalCode": "95014-2083",
      "contactStateProvince": "CA",
      "contactCity": "Cupertino",
      "countryName": "United States of America",
      "entityInternationalName": "APPLE, INC.",
      "entityContactAddressLine": "One Apple Park Way",
      "entityContactCountryCode": "USA",
      "entityContactPostalCode": "95014-2083",
      "entityContactStateProvince": "CA",
      "entityContactCity": "Cupertino",
      "entityCountryName": "United States of America",
      "entityOverviewDesc": "Apple, Inc. engages in the design, manufacture, and sale of smartphones, personal computers, tablets, wearables and accessories, and ..."
      "entityStatus": "",
      "size": ""
    }
  ]
}
```

```

    "entitySize": "",
    "ticker": "AAPL",
    "peerGroupId1": "18b47ee6-8356-443b-a903-b7311cd572f6",
    "peerGroupId2": "dcdc95b9-0af0-438e-af65-6299cblfaf88",
    "peerGroupId3": "",
    "peerGroupId4": "",
    "peerGroupId5": "",
    "rank": "",
    "nationalId": [
      {
        "idName": "isin",
        "idValue": "US0378331005"
      },
      {
        "idName": "cusip",
        "idValue": "037833"
      },
      ...
    ],
    "hasFinancials": "Yes"
  },
  ...
  ...
],
"total": 10000
}

```

4.2 Batch Entity Search Endpoint

Endpoint: /entity/v1/mapping

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
queries	Yes	An object containing the companies identifiers. The currently supported identifiers are listed in "Appendix: Supported Identifiers"; the keys accepted by the json for the main ones are: - entityidentifierbvd- Identifier from Orbis - entityidentifierorbis - Identifier from Orbis - entityInternationalName - entityIdentifierTaxNumber - entityIdentifierEin - Employer Reference Number - isin - International Securities Identification Number - cusip - Committee on Uniform Securities Identification Procedures - lei - Legal Entity Identifier - pid - CreditEdge Identifier - ticker - Stock symbol Provide the keys and the related values.	N/A	See below

Sample JSON request

```
{
  "queries": [
    { "entityidentifierbvd": "TW23535744" },
    { "pid": "037833" },
    { "isin" : "KR7000270009" },
    { "cusip" : "594918" },
    { "lei" : "7LTFWZYICNSX8D621K86" }
  ]
}
```

Response

The API response lists the same parameters as the basic search for the list of companies that match your search.

Sample JSON response

```
{
  "entities": [
    {
      "entityId": "TW23535744",
      "pid": null,
      "identifierOrbis": "060847165",
      "identifierBvd": "TW23535744",
      "primaryIndustryNACE": "",
      "primaryIndustryNAICS": "",
      "primaryIndustryNDY": "N48",
      "primaryIndustrySIC": "",
      "primaryIndustryNDYDescription": "SECURITY BROKERS & DEALERS",
      "type": "Bank",
      "entityType": "Bank",
      "internationalName": "IBF Securities Co., Ltd.",
      "contactAddressLine": "15f Sec. 5, Nanjing E. Rd. No 188 Zhongshan Dist. ",
      "contactCountryCode": "TWN",
      "contactPostalCode": "104",
      "contactStateProvince": "",
      "contactCity": " Taipei City",
      "countryName": "Taiwan, China",
      "entityInternationalName": "IBF Securities Co., Ltd.",
      "entityContactAddressLine": "15f Sec. 5, Nanjing E. Rd. No 188 Zhongshan Dist. ",
      "entityContactCountryCode": "TWN",
      "entityContactPostalCode": "104",
      "entityContactStateProvince": "",
      "entityContactCity": " Taipei City",
      "entityCountryName": "Taiwan, China",
      "entityOverviewDesc": "...",
      "entityStatus": "",
      "size": "",
      "entitySize": "",
      "ticker": "000779",
      "peerGroupId1": "53e0ad3c-45f0-4e93-a25e-5173e6ae19dd",
      "peerGroupId2": "357ce86d-07cd-40ce-9cc8-b50bbb6207cf",
      "peerGroupId3": "9f9779ba-e7a9-455b-a875-46219c54c850",
      "peerGroupId4": "d46c8ead-c8ba-485e-925a-9fa31131dee4",
      "peerGroupId5": "c0ccc41e-781e-4ffd-860f-762d3b854567",
      "rank": "",
      "nationalId": [
        {
          "idName": "ticker",
          "idValue": "000779"
        },
        {
          "idName": "lei",
          "idValue": "254900WBR5U6K5S0GL70"
        }
      ]
    }
  ],
}
```

```

    "hasFinancials": "Yes"
  },
  {
    "entityId": "US942404110",
    "pid": "037833",
    "identifierOrbis": "000411117",
    "identifierBvd": "US942404110",
    "primaryIndustryNACE": "2630",
    "primaryIndustryNAICS": "334220",
    "primaryIndustryNDY": "N21",
    "primaryIndustrySIC": "3663",
    "primaryIndustryNDYDescription": "ELECTRONIC EQUIPMENT",
    "type": "Corporate",
    "entityType": "Corporate",
    "internationalName": "APPLE, INC.",
    "contactAddressLine": "One Apple Park Way",
    "contactCountryCode": "USA",
    "contactPostalCode": "95014-2083",
    "contactStateProvince": "CA",
    "contactCity": "Cupertino",
    "countryName": "United States of America",
    "entityInternationalName": "APPLE, INC.",
    "entityContactAddressLine": "One Apple Park Way",
    "entityContactCountryCode": "USA",
    "entityContactPostalCode": "95014-2083",
    "entityContactStateProvince": "CA",
    "entityContactCity": "Cupertino",
    "entityCountryName": "United States of America",
    "entityOverviewDesc": "...",
    "entityStatus": "",
    "size": "",
    "entitySize": "",
    "ticker": "AAPL",
    "peerGroupId1": "18b47ee6-8356-443b-a903-b7311cd572f6",
    "peerGroupId2": "dcdc95b9-0af0-438e-af65-6299cblfaf88",
    "peerGroupId3": "",
    "peerGroupId4": "",
    "peerGroupId5": "",
    "rank": "",
    "nationalId": [
      {
        "idName": "isin",
        "idValue": "US0378331005"
      },
      {
        "idName": "cusip",
        "idValue": "037833"
      },
      {
        "idName": "entityIdentifierCik",
        "idValue": "0000320193"
      },
      {
        "idName": "sedol",
        "idValue": "2046251"
      },
      {
        "idName": "lei",
        "idValue": "HWUPKROMPOU8FGXBT394"
      },
      {
        "idName": "figi",
        "idValue": "EQ0010169500001000"
      },
      {
        "idName": "entityIdentifierEin",
        "idValue": "94-2404110"
      },
    ],
  }

```

```
        "idName": "entityIdentifierTaxNumber",
        "idValue": "94-2404110"
      },
      {
        "idName": "ticker",
        "idValue": "AAPL"
      }
    ],
    "hasFinancials": "Yes"
  }
],
"total": 0
}
```


5 Retrieving PD Values

Available Endpoints

End Point	Method	Description
/entities/pds	POST	Access our best estimate PD for a given ID. It is recommended that you use this endpoint unless you are an advanced user or use an existing product.
/entities/pds/creditedge	POST	Access CreditEdge PD values directly. Only listed companies appear since the model relies on equity market prices.
/entities/pds/riskcalc	POST	Access our best estimate PD from RiskCalc directly.
/entities/pds/payment	POST	Access PDs generated by the Payment model. Payment PDs are only calculated for US companies with netSales < 500mln USD for which payment information is available. It is not currently possible to provide your own payment data to generate results.

For the URL, combine the base URL ([Accessing the EDF-X API](#)) with the endpoint.

Entity ids can be found using the Entity search endpoint ([Entity Search](#)).

5.1 Request Structure

Request Parameters

Name	Required?	Description	Preset Value	Example
entities	Yes	A list of entityIds. We support BvD IDs and CreditEdge PIDs and the API automatically determines which one is being used.		"AT9110116332"
startDate	No	If provided, generates a history of PD values in the format YYYY-MM-DD.		"2021-01-01"
endDate	No	The date of the PD you are retrieving in the format YYYY-MM-DD.	Today's Date	"2021-06-09"

Name	Required?	Description	Preset Value	Example
historyFrequency	No	Defines the frequency of the PD value history. Allowed values are: "daily"; "monthly"; "quarterly"; or "annual". Note. Monthly time series are defined as follows. For public firms, the monthly time series is obtained from the daily series. The most representative data point is the last day of the month. Ex. the month of January is represented by the score at January 31st. For private firms, the monthly time series is the most granular currently available thus there is only one score per month. Ex. the months of January is represented by January score, that is marked with January 1st date. Note. Quarterly time series are obtained as follows. For public firms, the most representative data point is the last day of the quarter. Ex. Q1 is represented by March 31st score. For private firms, quarterly series are obtained from the monthly series. The most representative data point for a quarter corresponds to the first day of the following quarter e.g. for Q1 it would be the April monthly score.	"monthly"	"daily"
asyncResponse	No	Turns on asynchronous Mode.	false	false
asReported	No	Financial statements are usually not published on their statement date, they take a few months to finalize and file. There is a delay in new financial statements being available for the PDs calculations. During this period, EDF-X uses an older financial statement and adjusts the scores monthly using Credit Cycle Adjustments. When the new financial statement is available, EDF-X calculates PDs back to the financial statement date. This flag allows access to the PDs values as reported before they were reinstated.	false	false
modelParameters	No	Provides extra options for specific model features.		
fso				
includeDetail	No	Provides more information regarding the results.	false	false
- resultDetail - inputDetail - modelDetail				
processId	No	This parameter is alternative to providing the list of entities. The values accepted are process IDs received when uploading a csv file through the /modelInput endpoint.		43ef7678-fffb-414f-b151-c7225f325b03

Note When no startDate and endDate is provided, the API serves the latest PD available considered. For private firms this is the latest available month, while for active public firms this is the latest available day within the last 10 days. Dates beyond these can be accessed in the history by specifying startDate and endDate.

Sample JSON request

```
{
  "startDate": "2016-02-01",
  "endDate": "2018-03-01",
  "historyFrequency": "monthly",
  "asyncResponse": false,
  "asReported": false,
  "modelParameters": {
    "fso": false
  },
  "includeDetail": {
    "resultDetail": false,
    "inputDetail": false,
    "modelDetail": false
  },
  "entities": [
    {
      "entityId": "AT9110116332"
    }
  ]
}
```

5.1.1 Enabling Further Detail

The response provides the PD and associated information for the date range provided. However, if you want to view more specific information about the latest PD (as of the endDate) you can change the individual items within the includeDetail object literal to be true. Information is provided for the following areas:

- inputDetail - List of all inputs provided to model.
- resultDetail - List of all model outputs.
- modelDetail - Specifics on which model was used.

5.1.2 Using Model Parameters for Advanced Users

The Credit Cycle Adjusted (CCA) PD is returned automatically. To request the Financial Statement Only (FSO) PD instead set the "fso" flag to true in the "modelParameters" list.

```
{
  ....
  "modelParameters": {
    "fso": true
  },
  ....
}
```

5.1.3 Using in Asynchronous Mode

If the number of entities you are retrieving is large (>10000), it might be easier to use the asynchronous process. Enable by setting the asyncResponse parameter to true. The response returned is a processId. You can use asynchronous mode along

with processes endpoints to access the status and results after the request has been processed. For more details, see the [Accessing Processes](#).

```
{
  ...
  "asyncResponse": false,
  ...
},
....
}
```

5.1.4 Using processId

If you want to retrieve PDs for all the entities and financial statements that you uploaded via /modelInput.

```
{
  "startDate": "2016-02-01",
  "endDate": "2018-03-01",
  "historyFrequency": "monthly",
  "asyncResponse": false,
  "asReported": false,
  "modelParameters": {
    "fso": false
  },
  "includeDetail": {
    "resultDetail": false,
    "inputDetail": false,
    "modelDetail": false
  },
  "processId": "43ef7678-ffffb-414f-b151-c7225f325b03"
}
```

5.2 Response Structure

5.2.1 Response Sections

The following response sections are provided in the response under different conditions.

- PD Detail
 - This section is always returned and gives the user an overview of the PD as of the endDate provided.
- History
 - This section provides a PD time series for the date range requested.
 - Only provided if a start date that is less than the end date is part of the request.
- Model Detail
 - This section provides details about the model used.

- Only provided if includeDetail - modelDetail is True.
- Input Data
 - This section provides details about the inputs used
 - Only provided if includeDetail - inputDetail is True.
- Result Data
 - This section provides detailed model results
 - Only provided if includeDetail - resultDetail is True.

For full details of the outputs see the Appendix on Output Details.

Sample JSON Response

```
{
  "entities": [
    {
      "entityId": "AT9110116332",
      "asOfDate": "2018-03-01",
      "pd": 0.00289,
      "impliedRating": "Aa1",
      "confidence": "PF-G-S",
      "termStructure": {
        "forward": {
          "forwardly": "0.00289",
          "...",
          "forward9y": "0.005216"
        },
        "annualized": {
          "annualizedly": "0.00289",
          "...",
          "annualized10y": "0.004757"
        },
        "cumulative": {
          "cumulatively": "0.00289",
          "...",
          "cumulative10y": "0.046565"
        },
        "impliedRating": {
          "impliedRatingly": "Baa1",
          "...",
          "impliedRating10y": "Baa2"
        }
      },
      "history": [
        {
          "asOfDate": "2018-01-01",
          "pd": 0.002232,
          "impliedRating": "Aa1",
          "confidence": "PF-G-S",
          "termStructure": {
            "forward": {
              "forwardly": "0.002232", ...
            },
            "...",
            ...
          }
        },
        ...
      ]
    }
  ]
}
```

```

        "cumulative": {
            ...,
            "cumulative10y": "0.034667"
        }
        ...,
    },
    {
        "asOfDate": "2018-02-01",
        "pd": 0.002165,
        "impliedRating": "Aa1",
        "confidence": "PF-G-S",
        "termStructure": {
            "forward": {
                "forwardly": "0.002165",...
            },
            ...,
            "cumulative": {
                ...,
                "cumulative10y": "0.034667"
            }
            ...,
        }
    }
],
"modelDetails": {
    "modelId": "AUT40",
    "documentation": {
        "documentName": "linktodocument",
        "link": "documentTitle"
    }
},
"inputData": {
    "Cash": "62771.815",
    ...,
    "Profit Before Tax and Extra Items": "152471.463"
},
"resultData": {
    "Ratio: Size": "21.08884164176",
    "%-tile: Size": "1",
    "Forward 1-Yr. EDF": "0.00323279649",
    ...,
    "Ratio: Net Debt to Sales": "2.97963378471",
    ...,
    "Rel. Contrib. 5 Yr: Cash & Marketable Securities to Total Assets": "-0.12594671188"
}
}
]
}

```

Sample JSON response in Asynchronous Mode, or when the response is larger than around 1 MB

```

{
    "processId": "ed0ca90b-2b32-4a1a-9bbb-9571a7eadbc8"
}

```

6 Uploading Company Information

Available Endpoints

End Point	Method	Description
/entities/modelInputs	POST	Send your own data to our calculators.
/entities/financials/template/universal	GET	Access the csv file valid as universal input template.
/entities/financials/template/bank	GET	Access the csv file valid as input template for banks financial statements.

Use the /entities/modelInputs endpoint to send your own data to our calculators.

It is possible to calculate PDs, trade credit limits, etc., using company information you supply. The data is uploaded as a CSV file following a standard template. Two financial statement templates are supported, one specifically for assessing banks and a universal one for all other non-bank entities. Full financial statements are not required and only a minimal amount of information is required to generate a benchmark PD. Full details on the templates can be found in the [Input Fields](#).

The API responds with a processId, which can be used in the following ways to check the process status and retrieve the value.

1. With the /processes/{processId}/status endpoint to access the status of the upload, when they are ready the processId can be used as per point 2 and 3 below (See [Accessing Processes](#) for more information.).
2. In the /pds, /pds/riskcalc, /pds/creditedge endpoints as value for the parameter processId (See [Retrieving PD values](#) page for more information). Using it to access the PD, implied ratings, etc. for all the companies in the uploaded file.
3. By combining the entityIdentifier provided as input with the process id as follows: "{entityIdentifier}-{processId}". Using it as the identifier for the PD, trade credit limits, etc., endpoints to access the metrics calculated using the information uploaded for a subset of the companies in the uploaded file (See [Retrieving PD values](#) page for more information).

Note processIds are not permanent and uploaded data can be cleaned out as often as once a day.

6.1 Download the template

Endpoint: /entities/financials/template/universal

Method: GET

Request Parameters: None

Response structure: universal template CSV file, ready for download. For further description of the headers, see [Input Fields](#).

Endpoint: /entities/financials/template/bank

Method: GET

Request Parameters: None

Response structure: banks template empty CSV file, ready for download. For further description of the headers, see [Input Fields](#).

6.2 Upload the financial statements

Endpoint: /entities/modelInputs

Method: POST

For the URL, combine the base URL with the endpoint:

Request Body:

Type: form-data

Request Parameters

Name	Required?	Description	Preset Value	Example
uploadedFile	Yes	The template file containing the financial data		<template_file.csv>

Sample JSON response

```
{
  "processId": "ed0ca90b-2b32-4a1a-9bbb-9571a7eadbc8"
}
```


7 Accessing Processes

Use these endpoints to monitor and manage processes set up when running in asynchronous mode (parameter `asyncResponse = true`).

The `/status` endpoint can also be used to confirm the status of file uploading, using the `processId` received after the csv upload through `/modelInputs` endpoint. This is not true for the `/files` endpoint.

7.1 Status

Endpoint: `/processes/{processId}/status`

Method: GET

Type: Path parameter

Request Parameters

Name	Required?	Description	Preset Value	Example
processId	Yes	The process id of the process you want to retrieve the status of. The process id used can come from - endpoints where <code>asyncResponse</code> parameter is available ex. the endpoints to retrieve PDs i.e. <code>/pds</code>, <code>/pds/riskcalc</code>, <code>/pds/creditedge</code>; - the <code>/modelInputs</code> endpoint to check the status of the data upload.		7329f487-c110-47ff-a9a7-7a7bdf20f557

Response

The response can contain the following statuses:

- Initiated
- In progress
- Completed
- Errored

Sample JSON response

```
{
  "status": "Completed"
}
```

7.2 Files

Endpoint: /processes/{processId}/files

Method: GET

Type: Path parameter

Request Parameters

Name	Required?	Description	Preset Value	Example
processId	Yes	The process id of the process you want to retrieve the files from. The process id used can come from endpoints where asyncResponse parameter is available ex. the endpoints to retrieve PDs i.e. /pds, /pds/riskcalc, /pds/creditedge.		7329f487-c110-47ff-a9a7-7a7bdf20f557

Response

The JSON format response will contain a download link URL. A GET request with this URL will return the results for the asynchronous request in JSON format. The response from the download link will have the same structure as the equivalent synchronous data request.

Sample JSON response

```
{
  "downloadLink": "https://uswe2-XXX.s3.amazonaws.com/batch/755b5f53-86f6-4b9f-95c2-86597933b2d8_pdsdata.json?response-content-disposition=attachment&..."
}
```

8 Retrieving Limits for Trade Credit

Use the /tools/creditLimit endpoint to access estimated limits for trade credit. For each company except for financial institutions, we provide trade credit limits based on conservative, balanced, and aggressive risk appetites. Limits provide corporate users with guidance on how much trade credit to extend to a firm, and when to consider being paid upfront.

Limits are only implemented for entities that we have information on, either from Orbis or uploaded as custom financials. It is not possible to calculate limits based on custom PD inputs.

Endpoint: /tools/creditLimit

Method: POST

Request Parameters

Name	Required?	Description	Preset Value	Example
entities	Yes	A list of entityIDs. We support BvD IDs and CreditEdge PIDs and the API automatically determines which one is being used.		"AT9110116332"
startDate	No	If provided, generates a history of limit values to be generated in the format YYYY-MM-DD.		"2021-01-01"
endDate	No	The date of the limit you are retrieving in the format YYYY-MM-DD.	Today	"2021-06-09"

Sample JSON request

```
{
  "startDate": "2021-01-01",
  "endDate": "2021-06-09",
  "entities": [
    {
      "entityId": "AT9110116332"
    }
  ]
}
```

Response

For every month between startDate and endDate three values are provided corresponding to conservative, balanced, and aggressive assumptions, along with the PD used in the calculation.

Full details can be found in [Appendix: Output Details](#).

Sample JSON response

```
{
  "entityId": "US89238992",
  "creditLimits": [
    {
      "asOfDate": "2018-03-01",
      "pd": 0.02,
      "balanced": 23956.0,
      "aggressive": 29384723.0,
      "conservative": 239847234.0,
      "currency": "USD",
      "unit": 1
    },
    {
      "asOfDate": "2018-04-01",
      "pd": 0.02,
      "balanced": 23956.0,
      "aggressive": 29384723.0,
      "conservative": 239847234.0,
      "currency": "USD",
      "unit": 1
    }
  ],
}
```

9 Retrieving Peer Groups

The peer group endpoints allow the comparison of individual companies against their peers. These endpoints concern a set of pre-determined peer groups defined by geographic regions and/or industry segments; public and private companies are associated with one or more peer groups based on their traits. Based on this, it is possible to:

- view the risk of individual companies in the context of their peers by comparing the companies scores to the aggregations of the peer group;
- understand the trend of default at a macro level of geographic regions or industry segments.

Use these endpoints to retrieve the relevant information about pre-determined groups used for peer benchmarking.

There are four endpoints for this purpose:

Endpoint	Description
/entities/peers/id	For a given country and industry, it returns the relevant Peer Group ID or IDs.
/entities/peers/metrics	For a given peer group ID, you can explore the variable or variables (e.g. PD) that describe this peer group through to a list of available metrics.
/entities/peers/percentile	For a given peer group ID, you can obtain the percentile of a given value or level for the variable(s) that describe this peer group.
/entities/peers/metadata	Use this endpoint to retrieve descriptive information about a peer group.
/entity/peers/recommended	For given inputs, it returns only the relevant Peer Group ID that is also the Moody's Analytics recommended one.

For the URL, combine the base URL with the endpoint.

9.1 IDs

Use this endpoint to obtain the peer group ID for a given country and industry.

Endpoint: /entities/peers/id

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
peerRegion	Yes	Macro regions. Possible values: GLOBAL, AUSNZ (Oceania), CESOAM (Central and South Africa), EEURRUS (East Europe), MIDENOAF (Middle East and North Africa), NOASIA (North Asia), SEASIA (South East Asia), SOAM (South America), SOASIA (South Asia), USCAN (USA and Canada), WESTEUR (Western Europe). Either peerRegion or country need to be provided: provide peerRegion to obtain the regional peer groups.		WESTEUR
industryClassification	No	The industry classification for the primary industry. Possible value: NDY: Moody's Analytics private companies industry classification	NDY	NDY
industryCode	No	The primary industry code for the entity's economic activity. The code must be a valid code of the primaryIndustryClassification.		N09
ownershipType	Yes	Possible values: - For listed entities: "LISTED" - For not listed entities: "UNLISTED" "LISTED UNLISTED" If not provided, the results are generated for all three cases.		LISTED
country	No	Standard (ISO) three letter country codes. Either peerRegion or country need to be provided: provide country to obtain the country level peer groups.		DEU

See [Peer Groups - country to region mapping](#) for more information

Sample JSON request

```
{
  "peerRegion": "WESTEUR",
  "industryClassification": "NDY",
  "industryCode": "N55",
  "ownershipType": "UNLISTED"
}
```

Sample JSON response

```
{
  "results": [
    {
      "peerId": "4c41025d-6fc2-49c2-944d-e3b30b49c8d8"
    }
  ]
}
```

9.2 Metrics

Use this endpoint to retrieve metrics about variables (e.g. PD) within the peer group.

Endpoint: /entities/peers/metrics

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
peerId	Yes	The ID of the peer group for which you want to check the variable(s).		278a17e6-d69b-45a4-861a-ac4e20987436
variables	No	List of variables that describe the peer group and that you want to retrieve information about. See below the possible variables and associated metrics that are available for each peer group. If not defined, the default is annualizedcumulativepd1y.	annualizedcumulativepd1y	annualizedcumulativepd1y
metrics	Yes	List of metrics returned for the variable selected. Possible metrics are - Counter. - Percentiles. Available percentiles are: 25th (P025), 50th (P050) and all percentiles from 60th to 99th (P060...P099)		counter or P050

Variables and Metrics available

Variables	Metrics		Definition
	Counter	Percentiles	
annualizedcumulativepd1y	x	x	This variable describes the 1y CCA PD for the entities in the peer group.

Sample JSON request

```
{
  "peerId": "278a17e6-d69b-45a4-861a-ac4e20987436",
  "variables": ["annualizedcumulativepd1y"],
  "metrics": ["P025", "P050"]
}
```

Sample JSON response

```
{
  "peerId": "278a17e6-d69b-45a4-861a-ac4e20987436",
  "results": [
    {
```

```

    "asOfDate": "2021-08-01",
    "variableName": "annualizedcumulativepd1y",
    "unit": "1.0",
    "currency": null,
    "metricList": [
      "P025",
      "P050"
    ],
    "values": [
      0.0106414796715435,
      0.0105025072430054
    ]
  }
}

```

9.3 Percentile

For a specific variable, use this endpoint to retrieve the percentile of a given value.

Endpoint: /entities/peers/percentile

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
peerId	Yes	The ID of the peer group for which you want to check the variable(s).		278a17e6-d69b-45a4-861a-ac4e20987436
variableName	Yes	List of variables whose values will be checked against the percentiles values. The variables for which this information is available: • annualizedcumulativepd1y		annualizedcumulativepd1y
value	Yes	List of the values for which you would like to obtain the percentile it belongs to within the specified peer group.		0.008

Sample JSON request

```

{
  "peerId": "278a17e6-d69b-45a4-861a-ac4e20987436",
  "request": {
    "variableName": ["annualizedcumulativepd1y"],
    "value": [0.023]
  }
}

```

Sample JSON response


```
{
  "peerId": "278a17e6-d69b-45a4-861a-ac4e20987436",
  "results": [
    {
      "variableName": "annualizedcumulativepdy",
      "percentile": "P099"
    }
  ]
}
```

9.4 Metadata

Use this endpoint to retrieve information about a peer group.

Endpoint: /entities/peers/metadata

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
peerId	Yes	The ID of the peer group for which you want to get information such as - the number of companies in the peer group - the country, in ISO code standard - the industry, in NDY classification - whether the peer groups concerns listed or not listed companies		278a17e6-d69b-45a4-861a-ac4e20987436

Sample JSON request

```
{
  "peerId": "4c41025d-6fc2-49c2-944d-e3b30b49c8d8"
}
```

Sample JSON response

```
{
  "peerId": "4c41025d-6fc2-49c2-944d-e3b30b49c8d8",
  "count": 202767,
  "country": "GBR",
  "industryCode": "N55",
  "ownershipType": "UNLISTED"
}
```

9.5 Recommended

Endpoint: /entities/peers/recommended

Method: POST

Request Parameters:

Same as /entity/peers/id

Sample JSON request

```
{
  "industryClassification": "NDY",
  "industryCode": "N17",
  "ownershipType": "UNLISTED",
  "country": "USA"
}
```

Response:

Access the peer group ID that Moody's Analytics recommends as most appropriate according to both the criteria in the request and some pre-set criteria. The pre-set criteria currently in place is:

- the recommended peer groups always include 10 or more constituents.

Given the Country, Industry and Ownership type in the request, this endpoint only returns the Country level peer group that fits the criteria above; else the Region level peer group is looked up, and if that is not fitting the criteria either then the Global-wide peer group is returned as the recommended one.

In many cases, this response might be the same as /entity/peer/id.

10 Retrieving Early Warning Category and Trigger

The `riskCategory` endpoint allows you to access an early warning category for a given company. This category is chosen based on changes in a company's risk, as indicated by its implied rating, and how it compares to its peers. We use a target PD percentile of the peer group to define a trigger level and determine if a company's PD is above this level. The trigger methodology uses a Bayesian-adjustment to leverage information from more broadly defined peer groups to control for situations when a granular peer group has insufficient data. Additionally, it includes a procyclical adjustment that takes into account the position of the industry in the credit cycle.

The risk category is chosen using the following logic, based on a given company's calculated PD and implied rating change

Risk Category	PD vs trigger	Implied rating change
"Severe"	PD above trigger	Stable or worsening
"High"	PD above trigger	Improving
"Medium"	PD at or below trigger	Worsening
"Low"	PD at or below trigger	Stable or improving

The triggers end point provides access to a timeseries of trigger values.

10.1 Risk Category

The purpose of this endpoint is to provide an early warning category and trigger for an entity. The methodology uses change in implied rating and comparison of the PD values with its peer group.

Endpoint: </tools/riskCategory>

Method: POST

Request Parameters:

Name	Required	Description	Preset Value	Example
entities	Yes	A list of either: - entityId. We support BvD IDs and CreditEdge PIDs and the API automatically determines which one is being used. - A dictionary of: - entityId: A unique string used by requester to identify the entity. - pd: A PD value for the as of date. PDs should be a number between 0 and 1. - pdPrev: A PD value for a previous date (optional). PDs should be a number between 0 and 1. - peerId: A valid peer group identifier. This can be generated using the /edfx/v1/entities/peers/id endpoint		<pre>{ "entityId": "AT9110116332"}, { "entityId": "myExample1" , "pd": 0.03, "pdprev":0.024, "peerId": "89a28088-7952-4b 15-8ff3- f963e36d7cfd"}}</pre>
asOfDate	No	Date of PD value in the format YYYY-MM-DD. If this is not provided, this will default to the 1st day of the current month.	current date	"2021-10-01"
prevAsOfDate	No	Date of previous PD value in the format YYYY-MM-DD. this is used to calculate the change in implied rating.	asOfDate - 1 year	"2020-10-01"
targetPercentile	No	Target percentile used for trigger level. Percentiles are represented as numbers i.e. the 63rd percentile is represented as 0.63. Values should be between 0.60 and 0.95 and given to two decimal place precision.	This depends on the peer group chosen. - Private company peer group 0.80 - Public financial company peer group 0.85 - Public non-financial company peer group 0.75	0.75

Sample JSON request

```
{
  "entities":[
    {
      "entityId":"myExample1",
      "pd":0.03,
      "pdPrev":0.024,
      "peerId":"89a28088-7952-4b15-8ff3-f963e36d7cfd"
    },
    {
      "entityId": "AT9110116332"
    }
  ],
  "prevAsOfDate": "2021-09-01",
  "targetPercentile":"0.80"
}
```

Response

The response will be a list of elements with the following format:

- entityId: As sent in the request

- asOfDate: As sent in the request
- prevAsOfDate: As used in defining the irChange. If a prevAsOfDate was provided in the request, the same is returned, else asOfDate - 1 year; if less than 1 year of history is available, prevAsOfDate corresponds to the earliest data point available within 1 year.
- irChange: The notch difference between implied ratings at the two dates provided.
- trigger: The peer-based trigger PD value based on the chosen percentile.
- pd: The PD value used to derive the risk category
- riskCategory: The resulting early warning category for the asOfDate
- distanceToTrigger: The distance from PD to trigger in log space, defined as $\ln(\text{pd}/\text{trigger})$

Sample JSON response

```
{ {"entityId": "AT9110116332": ,
  "asOfDate": "2021-10-01",
  "irChange": 4,
  "trigger": 0.032,
  "pd": 0.04,
  "riskCategory": "Severe",
  "distanceToTrigger": 0.223},
  {"entityId": "myExample1": ,
  "asOfDate": "2021-10-01",
  "irChange": 2,
  "trigger": 0.04,
  "pd": 0.02,
  "riskCategory": "Medium",
  "distanceToTrigger": 0.693}}
```

10.2 Triggers

The purpose of this endpoint is to provide a timeseries of triggers for a given peer group.

Endpoint: /tools/triggers

Method: POST

Request Parameters:

Name	Required	Description	Preset Value	Example
peerId	Yes	peerId: A valid peer group identifier. This can be generated using • The entity endpoint with a valid BvD IDs or CreditEdge PID • The /entities/peers/id endpoint using peer traits (country, industry, ownership type)		"89a28088-7952-4b15-8ff3-f963e36d7cfd"
endDate	No	Date of last data point in the format YYYY-MM-DD.	current date	"2021-10-01"

Name	Required	Description	Preset Value	Example
startDate	No	Date of first data point in the format YYYY-MM-DD.	asOfDate - 1 year	"2020-10-01"
targetPercentile	No	Target percentile used for trigger level. Percentiles are represented as numbers i.e. the 63rd percentile is represented as 0.63. Values should be between 0.60 and 0.95 and given to two decimal place precision.	This depends on the peer group chosen. - Private company peer group 0.80 - Public financial company peer group 0.85 - Public non-financial company peer group 0.75	0.75

Sample JSON request

```
{
  "peerId": "89a28088-7952-4b15-8ff3-f963e36d7cfd",
  "startDate": "2020-11-01",
  "endDate": "2021-01-01",
  "targetPercentile": 0.80
}
```

Response

The response will elements with the following format:

- peerId: the peer id supplied as input
- triggers: a list containing the following data points:
 - asOfDate: Date of analysis
 - trigger: Timeseries of trigger value for given targetPercentile. This value has been adjusted to take into account an industries position in the credit cycle.
 - unadjustedTrigger: Timeseries of the raw percentile trigger value for given targetPercentile.

The response will provide all data points available. For public companies the data will be daily and for private companies it will be typically one data points per month.

Sample JSON response

```
{ "peerId": "89a28088-7952-4b15-8ff3-f963e36d7cfd": ,
  triggers: {
    { "asOfDate": "2021-10-01",
      "trigger": 0.02,
      "unadjustedTrigger": 0.033
    },
    . . . . .,
    { "asOfDate": "2021-9-01",
      "trigger": 0.0189,
      "unadjustedTrigger": 0.031
    }
  }
}
```



11 Retrieving Financial Information and Ratios

Use the financials endpoints to access information about financial statements and key ratios. For entities we have information on, you can download standard formats via the statements and ratios endpoints. Ratios can also be calculated using the /ratios/calculate endpoint when statement information is supplied.

11.1 Statements

Access a summary of the financial statement for entities we have data on.

Endpoint: </entities/financials/statements>

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
entities	Yes	A list of entityIds. We support BvD IDs and CreditEdge PIDs and the API automatically determines which one is being used.		"IT02558850828"
startDate	No	If provided, generates a history of PD values in the format YYYY-MM-DD.		"2010-12-31"
endDate	No	The date of the PD you are retrieving in the format YYYY-MM-DD.	Today's Date	"2022-03-01"

Sample JSON request

```
{
  "startDate": "2010-12-31",
  "endDate": "2022-03-01",
  "entities": [
    {
      "entityId": "IT02558850828"
    }
  ]
}
```

Response

The response will be a list of elements with the following format:

- **entityId:** As sent in the request
- **statement:** A list of statement objects with the following format:
 - **financialStatementDate:** Financial statement date
 - **consolidationCode:** Indicates is the statement is consolidated or not.
 - **filingType:** Indicates the filing type of the statement
 - **currency:** ISO currency code for the balance sheet
 - **balanceSheet:** A list of balance sheet items. All figures are in thousands as per the currency
 - **incomeStatement:** A list of income statement items. All figures are in thousands as per the currency

```
{
  "entities": [
    {
      "entityId": "IT02558850828",
      "statements": [
        {
          "financialStatementDate": "2010-12-31",
          "consolidationCode": "UNCONSOLIDATED",
          "filingType": "Local GAAP",
          "currency": "EUR",
          "balanceSheet": {
            "cashAndMarketableSecurities": 51.721,
            "totalAccountsReceivable": 59.77,
            "totalInventory": 47.7,
            "totalCurrentAssets": 161.846,
            "totalFixedAssets": 15.773,
            "totalIntangibleAssets": 0,
            "totalAssets": 177.619,
            "shortTermDebt": 0,
            "totalAccountsPayable": 13.712,
            "totalCurrentLiabilities": 33.565,
            "totalLongTermDebt": 0,
            "totalNonCurrentLiabilities": 132.528,
            "totalProvisions": 0,
            "totalLiabilities": 166.093,
            "netWorth": 11.526
          },
          "incomeStatement": {
            "netSales": 257.909,
            "totalWageExpense": 141.853,
            "totalAmortizationAndDepreciation": 5.059,
            "totalOperatingProfit": 11.856,
            "financeCosts": 0.542,
            "profitBeforeTaxesAndExtraordinaryExpenses": 11.315,
            "currentIncomeTaxExpense": 6.3,
            "totalExtraordinaryItems": -0.539,
            "netIncome": 4.476,
            "ebitda": 16.915
          }
        },
        ....
      ]
    }
  ]
}
```

11.2 Ratios

Access key financial ratios for entities we have data on.

Endpoint: /entities/financials/ratios

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
entities	Yes	A list of entityIds. We support BvD IDs and CreditEdge PIDs and the API automatically determines which one is being used.		"IT02558850828"
startDate	No	If provided, generates a history of PD values in the format YYYY-MM-DD.		"2010-12-31"
endDate	No	The date of the PD you are retrieving in the format YYYY-MM-DD.	Today's Date	"2022-03-01"

Sample JSON request

```
{
  "startDate": "2010-12-31",
  "endDate": "2022-03-01",
  "entities": [
    {
      "entityId": "IT02558850828"
    }
  ]
}
```

Response

The response will be a list of elements with the following format:

- entityId: As sent in the request
- ratios: A list of statement objects with the following format:
 - financialStatementDate: Financial statement date
 - leverage: A list of key leverage ratios
 - liquidity: A list of key liquidity ratios
 - operational: A list of key operations ratios
 - profitability: A list of key profitability ratios

Sample JSON response

```
{
  "entities": [
```

```

{
  "entityId": "IT02558850828",
  "ratios": [
    {
      "financialStatementDate": "2010-12-31",
      "leverage": {
        "ratioTotalDebtToTotalAssets": 0,
        "ratioTotalLiabilitiesToNetWorth": 14.4103,
        "ratioTotalLiabilitiesTotalAssets": 0.9351
      },
      "liquidity": {
        "ratioCurrentRatio": 4.8219,
        "ratioQuickRatio": 3.4007
      },
      "operational": {
        "ratioEBITDAToFinanceCosts": 31.2085,
        "ratioAssetTurnover": 2.9041
      },
      "profitability": {
        "ratioEBITToNetSales": 0.046,
        "ratioEBITDAToNetSales": 0.0656,
        "ratioNetIncomeToNetSales": 0.0174,
        "ratioReturnOnAssets": 0.0252,
        "ratioReturnOnAssetsBeforeTaxes": 0.0637,
        "ratioReturnOnCapital": 0.0823,
        "ratioReturnOnEquity": 0.3883,
        "ratioReturnOnEquityBeforeTax": 0.9817
      }
    },
    {
      "financialStatementDate": "2011-12-31",
      .....
    }
  ]
}

```

List of Ratios

Category	Ratio	Key	Ratio Definition
Leverage	Total Debt to Total Assets	ratioTotalDebtToTotalAssets	(shorttermdebt+ totallongtermdebt)/totalassets
Leverage	Total Liabilities to Shareholder's Equity	ratioTotalLiabilitiesToNetWorth	totalliabilities/networth
Leverage	Total Liabilities to Total Assets	ratioTotalLiabilitiesTotalAssets	totalliabilities/totalassets
Liquidity	Current Ratio	ratioCurrentRatio	totalCurrentAssets/totalCurrentLiabilities
Liquidity	Quick Ratio	ratioQuickRatio	(totalCurrentAssets - totalInventory)/totalCurrentLiabilities
Operational	Interest Coverage	ratioEBITDAToFinanceCosts	ebitda / financeCosts
Operational	Inventory Turnover	ratioInventoryTurnover	totalCostOfGoodsSold / ((totalInventoryPreviousYear + totalInventory)/2)
Operational	Total Asset Turnover	ratioAssetTurnover	netSales / ((totalAssetsPreviousYear + totalAssets)/2)
Profitability	EBIT Margin	ratioEBITToNetSales	totalOperatingProfit / netSales

Profitability	EBITDA Margin	ratioEBITDAToNetSales	ebitda / netSales
Profitability	Gross Profit Margin	ratioGrossIncomeToNetSales	grossIncome / netSales
Profitability	Net Profit Margin	ratioNetIncomeToNetSales	netIncome / netSales
Profitability	Return on Assets	ratioReturnOnAssets	netIncome / totalAssets
Profitability	Return on Assets before Tax	ratioReturnOnAssetsBeforeTaxes	profitBeforeTaxesAndExtraordinaryExpenses / totalAssets
Profitability	Return on Capital	ratioReturnOnCapital	totalOperatingProfit / (totalAssets - totalCurrentLiabilities)
Profitability	Return on Equity	ratioReturnOnEquity	netIncome / netWorth
Profitability	Return on Equity before Tax	ratioReturnOnEquityBeforeTax	profitBeforeTaxesAndExtraordinaryExpenses / netWorth

11.3 Ratio Calculation

Calculate key ratios using your own financial statement inputs

Endpoint: </entities/financials/ratios/calculate>

Method: POST

Request Parameters:

Name	Required?	Description	Preset Value	Example
statements	Yes	A list of statements. Each statement is a list of financial statement items - financialStatementDate - ebitda - financeCosts - grossIncome - netIncome - netSales - netWorth - profitBeforeTaxesAndExtraordinaryExpenses - shortTermDebt - totalAssets - totalAssetsPreviousYear - totalCostOfGoodsSold - totalCurrentAssets - totalCurrentLiabilities - totalInventory - totalInventoryPreviousYear	NA	See sample request

Name	Required?	Description	Preset Value	Example
		- totalLiabilities - totalLongTermDebt All items are optional. Missing values should not be sent and resulting ratios will be calculated as NA		

Sample JSON request

```
{
  "statements": [
    {
      "ebitda": "16.915",
      "financeCosts": "0.542",
      "grossIncome": "0",
      "netIncome": "4.476",
      "netSales": "257.909",
      "netWorth": "11.526",
      "profitBeforeTaxesAndExtraordinaryExpenses": "11.315",
      "shortTermDebt": "0",
      "totalAssets": "177.619",
      "totalAssetsPreviousYear": "0",
      "totalCostOfGoodsSold": "0",
      "totalCurrentAssets": "161.846",
      "totalCurrentLiabilities": "33.565",
      "totalInventory": "47.7",
      "totalInventoryPreviousYear": "0",
      "totalLiabilities": "166.093",
      "totalLongTermDebt": "0",
      "totalOperatingProfit": "11.856"
    }
  ]
}
```

Response

The response will be a list of elements with the following format:

- ratios: A list of statement objects with the following format:
 - financialStatementDate: Financial statement date
 - leverage: A list of key leverage ratios
 - liquidity: A list of key liquidity ratios
 - operational: A list of key operations ratios
 - profitability: A list of key profitability ratios

```
{
  "ratios": [
    {
      "financialStatementDate": "2010-12-31",
      "leverage": {
        "ratioTotalDebtToTotalAssets": 0,
        "ratioTotalLiabilitiesToNetWorth": 14.4103,
        "ratioTotalLiabilitiesTotalAssets": 0.9351
      },
      "liquidity": {
```

```
    "ratioCurrentRatio": 4.8219,  
    "ratioQuickRatio": 3.4007  
  },  
  "operational": {  
    "ratioEBITDAToFinanceCosts": 31.2085,  
    "ratioAssetTurnover": 2.9041  
  },  
  "profitability": {  
    "ratioEBITToNetSales": 0.046,  
    "ratioEBITDAToNetSales": 0.0656,  
    "ratioNetIncomeToNetSales": 0.0174,  
    "ratioReturnOnAssets": 0.0252,  
    "ratioReturnOnAssetsBeforeTaxes": 0.0637,  
    "ratioReturnOnCapital": 0.0823,  
    "ratioReturnOnEquity": 0.3883,  
    "ratioReturnOnEquityBeforeTax": 0.9817  
  }  
}  
]
```


12 Frequently Asked Questions

This list of Frequently Asked Question collects some useful information for the users to start understanding EDF-X. The questions are grouped in the following topics:

- Data
- Scoring Process
- Results

12.1 Data

How frequently is the data updated?

The PDs and all the related analytics are updated daily for most listed companies and monthly for all others.

How can I access PD values - do I have to map and upload financial statement data?

For companies in Orbis no data needs to be uploaded - the full Orbis universe is available and accessible through the entity's BVDID. Additionally, many listed company PDs can be accessed using the CreditEdge PID.

Financial Statements can still be uploaded for other companies, or where the user wants to use their own data and the PD is calculated. **Are the PD resulting from EDF-X the same**

How can I create the input template to upload my financial statements?

To create the input template: create a CSV file where the column headers match the column names in the "Appendix: Inputs Details" section. When filling the template, it is important to review the dates formats and make sure that the mandatory fields are correct.

Is there a specific input template for entities in different countries?

No. There are two input templates supported – one for banks and one for all non-bank corporates – and they are consistent globally.

Is any currency conversion applied to input data?

Currency conversion is applied when the PD calculation methodology requires it; the exchange rates are updated monthly. Currency conversion is not needed when the financial statements are already in the appropriate currency.

For some entities, more than a financial statement is available in Orbis: which one is used to feed the PD calculations?

The selection of the financial statement considers the Consolidation code, which is the variable detailing what type of statements are available. The priority is assigned following this order:

1. Statement of a mother company integrating the statements of its controlled subsidiaries or branches with an unconsolidated companion (for Orbis users, this corresponds to code C2)
2. Statement of a mother company integrating the statements of its controlled subsidiaries or branches with no unconsolidated companion (corresponds to code C1)
3. Additional consolidated statement, if available (corresponds to code CA2)
4. Statement not integrating the statements of the possible controlled subsidiaries or branches of the concerned company with a consolidated companion (corresponds to code U2)
5. Statement not integrating the statements of the possible controlled subsidiaries or branches of the concerned company with no consolidated companion (corresponds to code U1)
6. Additional unconsolidated statement, if available (corresponds to code UA2)
7. Limited financial data available (corresponds to code LF).

Currently, only financial statements that are 12 months apart are used in the scoring pipeline.

Is it possible to upload more than one financial statement for the same company?

Yes, as long as they have different financial statement dates. You can upload more than one financial statement for the same company by populating two different rows of the same input CSV files. Use the same entity identifier and company information (fields in blue in "Appendix: Inputs Details") but specifying the different statement dates and financial statements information (fields in yellow in "[Appendix: Inputs Details](#)").

Does the input data go through quality checks?

Yes, all inputs are checked against a list of business rules that ensure the input is of good quality. In case a financial statement violates some of the business rules, a flag highlights the breach but the data is used in calculations; no financial statement is blocked or dropped due to quality check violations. The outcome of the checks on the input data is also presented in the Confidence indicator in the API response (for more details see the description of Confidence in [Appendix: Output Details](#)).

There are two kinds of rules enforced.

Balance Rules

Rule	Description
Balance Sheet	$\text{totalAssets} = \text{totalLiabilities} + \text{netWorth}$
Total Assets	$\text{totalAssets} = \text{totalCurrentAssets} + \text{totalFixedAssets}$
Total Liabilities	$\text{totalLiabilities} = \text{totalCurrentLiabilities} + \text{totalNonCurrentLiabilities}$
Total Fixed Assets	$\text{totalFixedAssets} > \text{totalIntangibleAssets}$
Profit Before Tax and Extra	$\text{profitBeforeTaxesAndExtraordinaryExpenses} > \text{totalOperatingProfit} - \text{financeCosts}$
Total Current Assets	$\text{totalCurrentAssets} > \text{totalInventory} + \text{totalAccountsReceivable} + \text{cashAndMarketableSecurities}$

Rule	Description
Total Current Liabilities	$\text{totalCurrentLiabilities} > \text{totalAccountsPayable} + \text{shortTermDebt}$

Positive Checks

Rule	Description
Negative Total Assets	$\text{totalAssets} < 0$
Negative Total Liabilities	$\text{totalLiabilities} < 0$
Negative Total Current Assets	$\text{totalCurrentAssets} < 0$
Negative Total Current Liabilities	$\text{totalCurrentLiabilities} < 0$
Negative Cost of Goods Sold	$\text{totalCostOfGoodsSold} < 0$
Negative Finance Costs	$\text{financeCosts} < 0$
Negative Cash and Marketable Securities	$\text{cashAndMarketableSecurities} < 0$
Negative Accounts Receivable	$\text{totalAccountsReceivable} < 0$
Negative Inventory	$\text{totalInventory} < 0$
Negative Fixed Assets	$\text{totalFixedAssets} < 0$
Negative Current Debt	$\text{debtCurrentMaturities} < 0$
Negative Accounts Payable	$\text{totalAccountsPayable} < 0$
Negative Current Liabilities	$\text{totalCurrentLiabilities} < 0$
Negative Long Term Debt	$\text{totalLongTermDebt} < 0$
Negative Net Sales	$\text{netSales} < 0$
Negative Wage Expense	$\text{totalWageExpense} < 0$

The entity does not have an industry code, can a PD still be calculated?

Yes, the PD can still be calculated. If no industry code is available for the entity under analysis, we either use a proxy based on the entity type or score the entity as if the industry was unassigned, which covers both entities without industry information and those whose industry does not fit into our standard industry categories.

How is trade-payment data used in EDF-X?

The trade-payment data currently used in EDF-X is sourced from Moody's Analytics Pulse (previously known as Cortera); this data concerns small and medium unlisted firms in the USA, it is typically very recent and it is updated with monthly frequency in EDF-X. Trade-payment data feeds our RiskCalc Payment model, which calculates a PD without the need of financial statements information.

12.2 Scoring Process

How is the appropriate model selected?

The user is not required to select a model; the appropriate model is associated to the entities based on the country, size, industry, and type of entity.

Can a PD be produced with no financial statement?

Yes. We provide a benchmark PD based on country and industry combination, considering the company size and leverage where available.

Can a Benchmark PD be produced with no entity ID?

Yes. If the company is not available in Orbis, you can provide via upload the company information (See [Input Fields](#)) that is used to calculate the PD.

How is a PD calculated for entities in countries that are not directly supported by one of the RiskCalc models?

In these cases, the PD is calculated with the model associated to the entity, either a RiskCalc model or the Benchmark model, and adjusted with the appropriate country factor. This factor is calculated by considering country information like GDP, unemployment rate, sovereign ratings, and loan loss provision data. Thanks to the country factors, the PD calculated by the model is adjusted to reflect the specificities of the entity's country. The adjustment is only applied in addition to the calculated PD and it does not impact the ratios and drivers displayed in the result data section. The country factor is updated monthly, and the PD is adjusted with the country factor at the statement date. The outcome of these steps is the FSO PD. The CCA PD is also available, and calculated using this adjusted FSO PD as input.

How are the country adjustment factors provided to the process?

The user is not required to provide any country factor. The scoring process takes care of selecting the appropriate factor and it returns the PD already adjusted. The country factors coverage is global: the country factor is estimated monthly, for every country in the world.

Can the user provide custom country factors?

No, the user cannot input the country factors.

How is the PD adjusted for the credit cycle?

For non-listed firms, PD values are based on a financial statement and then adjusted monthly using a credit-cycle multiplier. The multiplier embeds country and industry-specific elements, and it references analytics derived from the CreditEdge model and other macro data. Listed firm use market data and are updated daily.

12.3 Results

What is the difference between FSO and CCA PD values?

Financial Statements Only (FSO) PD values are calculated starting from the information in the companies financial statements. Credit Cycle Adjusted (CCA) PD values adjust the FSO PD to reflect the impact of the credit cycle for the

specific country and industry of the entity under analysis. The CCA PD is a more point-in-time estimate as it includes a wider range of available information, which makes it more volatile than the FSO PD. The CCA PD values are provided by default. The "statement date" clarifies the date of the financial statement used in the analysis. Despite the unchanged financial statement for a number of months (therefore unchanged FSO PD), the CCA PD value changes because of the credit cycle adjustment.

What is the difference between the annualized, cumulative, and forward term structure?

See the description in the appropriate paragraph of [Appendix: Output Details](#) description.

Which PD is shown in the main part of the EDF-X response?

As described in "Output fields", the one year PD. This value is the same irrespective of the type of term structure.

How is the confidence indicator interpreted?

Confidence is an indicator that provides concise information about the elements playing a role in the PD calculations. For example, at a glance a the user can understand whether full financial statements were available, the age of the statements and their quality. This indicator is available in two formats: the code and the description. For more details about the possible values, see [Appendix: Output Details](#) or the summary tables below.

The Confidence Code has the form: ModelType-DataQuality-DataAge-Warnings e.g. PF-G-R-ACT_IMP, where warnings can be omitted if there are none.

The first table explains the code's core components; the second table describes the additional warnings that the code contains; finally, the last table shows a list examples of the possible codes. The order in the tables is not representative of the increasing/decreasing of confidence per se.

Confidence Code Components: Core Part

Code Component	Possible Values	Description
Model Type	P	Confidence code starting with "P" indicates that the PD was calculated using the CreditEdge Public firm model (for the version, see the Model Inventory page).
	PF	Confidence code starting with "PF" indicates that the PD was calculated using a Private firm model from the RiskCalc Model Suite, with Full financials available.
	PP	Confidence code starting with "PP" Indicates that the PD was calculated using a Private firm model from the RiskCalc Model Suite, with Partial financials available hence a missing information treatment is applied in the model.
	PY	Confidence code starting with "PY" Indicates that the PD was calculated applying the Payment model to trade-payment data from Moody's Analytics Pulse (Cortera) and no financial statements were available.
	PN	Confidence code starting with "PN" Indicates that the PD was calculated using the Benchmark model because No financial information is available or the latest informational available is too old hence cannot be considered relevant.
Data Quality	G	The financial statement passed all the quality checks (listed above in this F.A.Q.).

Code Component	Possible Values	Description
Data Age	E	Some of the items in the input financial statements use estimates by the information provide, rather than the actual values.
	I	The input financial statement failed some of the quality checks.
	X	This is a "placeholder" added when the information is not relevant.
	R	Financial statements are considered Recent if the Statement date is earlier than 18M (included).
	O	Financial statements are considered Old if the Statement date is between 18M and 36M (included).
	S	Financial statements are considered Stale if the Statement date is older than 36M.
	X	This is a "placeholder" added when the information is not relevant.

Confidence Code Components: Additional Warnings

Code Component	Possible Values	Description
Legal Status		
	BKRPT	This code replaces any other component, and it flags that the entity is Bankrupt.
	ACT_IMP	This code is chained to the confidence code when the entity is active but some distress occurred hence the legal status is Active (default of payment), Active (rescue plan), or Active (insolvency proceedings).
	NOT_ACT	This code is chained to the previously described component when the entity is not active. It includes cases when then legal status is Inactive (no precision), In Liquidation or Dissolved (any form of dissolved).
Other		
		Additional flags can be chained to the code to indicate • triggers where the PD is generated, but the specific type of company was not included in the model development sample • triggers based on industry assignment
	FLAG_L	Flags that the PD is calculated using a model that is not specific for the unlisted entity under analysis, which is identified, via the Legal Form, as Nonprofit Organizations or Public Limited Company
	FLAG_T	Flags that the PD is calculated using a model that is not specific for the unlisted entity under analysis, which is identified, via the Entity Type, as a Marine Vessels or a Public Authority, a State, a Government, a Foundation or a Research Institute.
	NN	Flags that there is no available information about the entity's industry
	NP	Flags that while there is no available information about the entity's industry, a proxy could be assigned thanks to available information about the Entity Type. This is possible when the entity has one of the following Entity Types: Bank; Mutual and Pension Fund; Nominee; Trust; Trustee; Financial Company; Insurance Company; Private Equity Firm; Venture Capital; Hedge Fund; Marine vessels; Foundation; Research Institute.

Confidence Code: Examples

Code	Description
BKRPT	Entity is bankrupt.
P-G-R	Public firm, CreditEdge model. Based on recent and good quality inputs.

Code	Description
PF-G-R	Private firm, RiskCalc model with full financials available, recent and good quality inputs.
PF-G-R-NOT_ACT	Private firm, RiskCalc model with full financials available, recent and good quality inputs. Entity is not active (in liquidation, dissolved or inactive).
PF-E-R	Private firm, RiskCalc model with full financials available, recent with some estimated inputs.
PP-I-O	Private firm, RiskCalc model with partial financials available, less recent and some inputs failed validation rules.
PP-I-O-ACT_IMP	Private firm, RiskCalc model with partial financials available, less recent and some inputs failed validation rules. Entity is active but status is impaired (insolvent, in default or rescue plan).
PY-X-R	Private firm, Payment model with recent information.
PN-X-X-FLAG_L	Private firm benchmark model with no financials available. Legal form was not included in the model development sample.
PN-X-X-FLAG_L-NN	Private firm benchmark model with no financials available. Legal form was not included in the model development sample. Industry not available.
PN-X-X-FLAG_T-NR	Private firm benchmark model with no financials available. Entity type was not included in the model development sample. Proxy industry inferred from entity type.
PN-X-X	Private firm benchmark model with no financials available.
PN-X-S	Private firm, RiskCalc model with outdated financial statements.

How is the implied rating derived?

The implied rating is derived from the calculated PD using a lookup which assigns PD ranges to individual Moody's ratings. This mapping is designed to be reasonably consistent with the default rates of bond ratings as measured by Moody's Default Studies. A static mapping derived from long-run defaults is used for unlisted entities. For listed companies, a dynamic mapping is used. This is derived by considering median PDs of rated entities for each rating bucket and updated monthly.

Do the results data go through quality checks?

The scoring system goes through several level of rigorous testing before we publish it to production. In addition, automated checks have been implemented to validate the completeness of results and the reasonability of each PD produced.

Are the PD resulting from EDF-X the same as those coming from RiskCalc and CreditEdge?

In some cases they will. For listed entities, the PD from EDF-X will be the same as the PD from the CreditEdge EDF9 model. For unlisted companies, EDF-X uses the models from the RiskCalc Models Suite, although some differences may be found between the results in RiskCalc and the results in EDF-X.

- For entities scored with the Emerging Markets Model (EMM3.1): RiskCalc applies a static country adjustment, and it is up to the users to specify their own adjustments; EDF-X uses new country adjustment factors, estimated monthly and applied automatically for each statement;
- EDF-X only supports the latest available version of the RiskCalc models, hence an entity that RiskCalc scores using an old model ex. Germany 3.2, might have a different PD in EDF-X because the new model ex. Germany 4.0 is applied instead;

- In cases where pre-populated financial statements are selected in RiskCalc, EDF-X could return a different result because of differences between data providers. In RiskCalc, some data providers substitute zero for missing information, while EDF-X will leave this information blank. In cases where the missing information is a mandatory input for the corresponding RiskCalc model, such as in the case of UK entities where the income statement is not always publicly available, EDF-X will provide a benchmark model result.
- EDF-X uses a universal template for financial statements import, while RiskCalc models have models specific template; some minimal impact on PD might occur depending on how data is mapped to these templates;
- For entities with no financial statements: EDF-X returns a score thanks to the Benchmark model.

Can I assess how a listed entity would be scored if no Public Entities Model was available?

Yes, It is possible to access the PD generated by the most appropriate RiskCalc model for listed entities too, by using the endpoint `/pds/riskcalc`. However, this is not the best score that we provide for listed entities. The best score for each entity can be accessed using the `/pds` endpoint.

Can I access the current score for outdated financial statements?

If the entity is part of Orbis, and only outdated financial statements are available there then you can access a score for the current date: the score will be calculated using the Benchmark model. If you are uploading your own financial statement, you can obtain the current score by populating the `asOfDate` in the input template with the date that you want to score the statement through.

How do you handle cases of formerly public companies?

For companies that used to be public and got delisted, EDF-X serves the PD calculated via CreditEdge public firms model up to the date when the company was delisted. No score is served after that date (consistently with CreditEdge).

13 Appendix: Input Details

If you have access to financial statements for an entity not available in the database or an entity that is available in the database but without financial statements, you can import statements and obtain a related PD. You can also access a PD when minimum information is available, including cases with no ID or financial statements. In these circumstances, you can upload the same template and fill in a minimum set of columns related to the entity's country and industry. If available, provide measures of the entity's size (Total Assets) and leverage (Total Liabilities and Total Assets) to improve the calculation.

The input fields detailed in the following tables are grouped in two categories.

- Company Information - Static information, highlighted blue. A minimal set of information alone can still be used to derive a PD .
- Financial Statement Information - Information, highlighted yellow, that changes at least each statement date. This information is not strictly required to calculate a score, however the more information provided, the more accurate the score is. We recommend you provide at least a measure of the Total Asset.

The input CSV file with the columns listed in the tables [can be uploaded](#). The upload must have all columns listed, even empty columns. [The name and order of columns must reflect the following table](#). Finally, the name of the CSV file uploaded cannot have white spaces, use an underscore instead.

We provide two templates - a universal one for all non-banks and the other specifically for banks. You need to upload each separately.

The company imported with these templates will always be assigned one of the RiskCalc models for private entities.

13.1 Universal Input Template - Minimal

A small number of field are needed to produce an PD using our benchmark model. The following table defines this minimal set. Note it is a subset of both the other templates.

Minimal Valid Template Upload Information

Columns name	Required	Field Type	Standard Values/Format	Definition
entityInternationalName	No	String		The international name for the entity.

Columns name	Required	Field Type	Standard Values/Format	Definition
entityIdentifierbvd	Yes	String		A unique identifier for each entity. This field must be populated.
financialStatementDate	Required if totalAssets or totalLiabilities provided	Date	YYYY-MM-DD	The closing date of the financial statement period.
primaryIndustryClassification	Yes	String	NDY, NACE2, NAICS2017	The industry classification for the primary industry. • NDY: Moody's Analytics private companies industry classification • NACE2 as per https://ec.europa.eu/eurostat/web/nace-rev2 • NAICS2017 as per https://www.census.gov/naics/
primaryIndustry	Yes	String		The primary industry code for the entity's economic activity. The code must be a valid code of the primaryIndustryClassification. When Orbis provides more than an industry code, the primary code is used.
primaryCountry	Yes	String		The primary country for the entity's economic activity. Valid values are the three-letter country codes from ISO (https://www.iso.org/iso-4217-currency-codes.html).
primaryStateProvince	Recommended in US only	String		The primary state or province for the entity's economic activity. Provide the subdivision ISO 3166 code (https://www.iso.org/iso-3166-country-codes.html) removing the country prefix. For example, the ISO 3166 code for California is US-CA, so simply use "CA" in the input file. Note the field is only currently used for entities in the United States of America.
currency	Required if totalAssets or totalLiabilities provided	String		The currency denomination. Valid values are ISO 4217 currency codes. All financial input fields must be expressed in the financial statement currency.
totalAssets	Recommended for more accurate results	Numeric	Unit: Thousands, in the specified currency	totalAssets represents all valuable resources owned or controlled by the entity that have been acquired at a measurable cost. It is the sum of current assets, fixed(tangible) assets, intangible assets, other non-current assets, less accumulated depreciation and amortization. This field is used as an indicator of the size of the entity. It is not mandatory however it leads to a more accurate PD measure if provided.
totalLiabilities	Recommended for more accurate results	Numeric	Unit: Thousands, in the specified currency	totalLiabilities represents all debt and other outstanding obligations at the close of the prior fiscal year, excluding total provision.

13.2 Universal Input Template - Full

Valid Template Upload Information for all non-banks

Columns name	Field Type	Standard Values/Format	Definition
entityInternationalName	String		The international name for the entity.
entityIdentifierbvd	String		A unique identifier for each entity. This field must be populated.
financialStatementDate	Date	YYYY-MM-DD	The closing date of the financial statement period.
asOfDate	Date	YYYY-MM-DD	The date when the data snapshot details are relevant. Currently not relevant for PD calculations.
primaryIndustryClassification	String	NDY, NACE2, NAICS2017	The industry classification for the primary industry. - NDY: Moody's Analytics private companies industry classification - NACE2 as per https://ec.europa.eu/eurostat/web/nace-rev2 - NAICS2017 as per https://www.census.gov/naics/
primaryIndustry	String		The primary industry code for the entity's economic activity. The code must be a valid code of the primaryIndustryClassification. When Orbis provides more than an industry code, the primary code is used.
primaryCountry	String		The primary country for the entity's economic activity. Valid values are the three-letter country codes from ISO (https://www.iso.org/iso-4217-currency-codes.html).
primaryStateProvince	String		The primary state or province for the entity's economic activity. Provide the subdivision ISO 3166 code (https://www.iso.org/iso-3166-country-codes.html) removing the country prefix. For example, the ISO 3166 code for California is US-CA, so simple input CA. Note the field is only currently used for entities in the United States of America.
currency	String		The currency denomination. Valid values are ISO 4217 currency codes. All financial input fields must be expressed in the financial statement currency.
entityLegalForm	String	BranchChamberOffice, C_Corporation, CollectiveEnterprise, Cooperative, Corporation, Farm, ForeignEntity, Government, Institution, JointStockCompany, JointVenture, LimitedLiabilityCompany, LimitedLiabilityPartnership, NonProfit, Other, Partnership, PublicLimitedCompany, S_Corporation, SoleProprietorship, TrustFinancialInstitution, Unknown	The standardized legal forms.
auditQuality	String		The audit opinion provided by an auditor of the financial statement.

Columns name	Field Type	Standard Values/Format	Definition
entityType	String	Private Firm, Public Firm, Sovereign, Municipality, and Supranational, Unknown.	Type that describes the entity.
entityStatus	String	Active, Active (default of payment), Active (dormant), Active (insolvency proceedings), Active (reorganization), Active (rescue plan), Bankruptcy, Dissolved, Dissolved (bankruptcy), Dissolved (demerger), Dissolved (liquidation), Dissolved (merger or take-over), In liquidation, Inactive (no precision), Unknown	The status of the entity.
entityStatusDate	Date	YYYY-MM-DD	Date that the entity changed to the current entity status.
cashAndMarketableSecurities	Numeric	Unit: Thousands, in the specified currency	The cashAndMarketableSecurities represents the total value of the entity's most liquid assets, comprised of cash, cash equivalents, and short-term investments at the end of the fiscal period. The components of Cash and Marketable Securities must be under current assets and do not include long-term investments.
cashAndMarketableSecuritiesPreviousYear	Numeric	Unit: Thousands, in the specified currency	The cashAndMarketableSecurities from the previous year of provided financialStatementDate.
totalAccountsReceivable	Numeric	Unit: Thousands, in the specified currency	The totalAccountsReceivable represents the entity's claims arising from the sale of goods and services to customers, on credit, that are expected to be settled in cash, within the current period.
totalAccountsReceivablePreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalAccountsReceivable from the previous year of provided financialStatementDate.
totalInventory	Numeric	Unit: Thousands, in the specified currency	The totalInventory represents the merchandise/goods bought for resale and materials and supplies purchased for use in production of revenue. It includes purchase cost, sales cost, sales taxes, transportation costs, insurance costs, and storage costs. Inventories are taken directly from the balance sheet, without any alterations for accounting method (for example, LIFO, FIFO, Avg. Cost). Also called, Stocks.
totalInventoryPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalInventory from the previous year of provided financialStatementDate.
totalCurrentAssets	Numeric	Unit: Thousands, in the specified currency	The totalCurrentAssets represents all assets that are expected to be used up or sold in the current fiscal period. It can be converted into cash within the normal operating business cycle and shows how much the entity has, if needed immediately, and how liquid the entity is. Usually include cash & marketable securities, inventory, accounts receivable, and other current assets.
totalCurrentAssetsPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalCurrentAssets from the previous year of provided financialStatementDate.
totalAccumulatedDepreciation	Numeric	Unit: Thousands, in the specified currency	totalAccumulatedDepreciation represents the cumulative depreciation expense recorded to allocate the cost of a tangible (depreciation) or intangible (amortization) asset over its useful life.
totalAccumulatedDepreciationPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalAccumulatedDepreciation from the previous year of provided financialStatementDate.

Columns name	Field Type	Standard Values/Format	Definition
totalFixedAssets	Numeric	Unit: Thousands, in the specified currency	The totalFixedAssets represents the total sum of all tangible assets that are meant for permanent ownership or use. Usually includes tangible accounts such as Land, Buildings, Machinery, Tools.
totalFixedAssetsPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalFixedAssets from the previous year of provided financialStatementDate.
totalIntangibleAssets	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The totalIntangibleAssets represents the value of all assets that are beneficial to companies but do not have any physical attributes. Examples include patents, licenses, trademarks, concessions, and goodwill.
totalAssets	Numeric	Unit: Thousands, in the specified currency	The totalAssets represents all valuable resources owned or controlled by the entity that have been acquired at a measurable cost. It is the sum of current assets, fixed(tangible) assets, intangible assets, other non-current assets, less accumulated depreciation and amortization. This field is used as an indicator of the size of the entity. It is not mandatory however it leads to a more accurate PD measure if provided.
totalAssetsPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalAssets from the previous year of provided financialStatementDate.
notesPayable	Numeric	Unit: Thousands, in the specified currency	The notesPayable represents the short-term (less than 12 months or one operating cycle) obligation from written promissory notes, including bank and commercial papers.
debtCurrentMaturities	Numeric	Unit: Thousands, in the specified currency	The debtCurrentMaturities represents the portion of the principal payments of long-term, interest-bearing obligations due within the next 12 months or one operating cycle.
shortTermDebt	Numeric	Unit: Thousands, in the specified currency	The shortTermDebt is the sum of all debts that dues within the next 12 months or one operating cycle. Includes notes payable, debt current maturities, and other short-term debts.
shortTermDebtPreviousYear	Numeric	Unit: Thousands, in the specified currency	The shortTermDebt from the previous year of provided financialStatementDate.
totalAccountsPayable	Numeric	Unit: Thousands, in the specified currency	The totalAccountsPayable represents the obligations arising from the firm's ongoing operations, including the purchase of goods, materials, supplies, and services used in the production and sale of goods or services that due within one year.
totalAccountsPayablePreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalAccountsPayable from the previous year of provided financialStatementDate.
totalCurrentLiabilities	Numeric	Unit: Thousands, in the specified currency	Obligations to be settled soon, typically within one year. Include short-term debt, accounts payable, and other current liabilities.
totalCurrentLiabilitiesPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalCurrentLiabilities from the previous year of provided financialStatementDate.
totalLongTermDebt	Numeric	Unit: Thousands, in the specified currency	This item represents all obligations such as bank loans or bonds or other forms of debt that are due after 12 months or one operating cycle.

Columns name	Field Type	Standard Values/Format	Definition
totalLongTermDebtPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalLongTermDebt from the previous year of provided financialStatementDate.
totalNonCurrentLiabilities	Numeric	Unit: Thousands, in the specified currency	totalNonCurrentLiabilities represents obligations listed on the balance sheet not due for more than 12 months or one operating cycle. Also known as long-term liabilities.
totalNonCurrentLiabilitiesPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalNonCurrentLiabilities from the previous year of provided financialStatementDate.
totalProvisions	Numeric	Unit: Thousands, in the specified currency	The totalProvisions represents the liabilities to cover expected future losses, which are either likely or certain to be incurred. Amounts can be reliably estimated as to the date when they arise.
totalLiabilities	Numeric	Unit: Thousands, in the specified currency	The totalLiabilities represents all debt and other outstanding obligations at the close of the prior fiscal year, excluding total provision.
totalLiabilitiesPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalLiabilities from the previous year of provided financialStatementDate.
retainedEarnings	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The retainedEarnings represents the cumulative earnings of the entity less total dividend distributions to shareholders. Typically, it is the prior year's retained earnings plus net income less distributions.
netWorth	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The netWorth represents the sum of all equity items, including retained earnings and other equity. Also known as total shareholder equity.
netSales	Numeric	Unit: Thousands, in the specified currency	The netSales represents the sum of the total (or gross) sales minus its returns, allowances, and discounts.
netSalesPreviousYear	Numeric	Unit: Thousands, in the specified currency	The netSales from the previous year of provided financialStatementDate.
totalCostOfGoodsSold	Numeric	Unit: Thousands, in the specified currency	The totalCostOfGoodsSold represents the total costs used to create a product or service, which has been sold. These costs include direct labor, materials, and overhead.
grossIncome	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The grossIncome represents the difference between net sales and cost of goods sold(COGS).
totalWageExpense	Numeric	Unit: Thousands, in the specified currency	The totalWageExpense represents the wage and salary expenses and other employee benefits.
totalAmortizationAndDepreciation	Numeric	Unit: Thousands, in the specified currency	The totalAmortizationAndDepreciation represents the expense recorded in current fiscal year to allocate the cost of a tangible asset (depreciation) or intangible asset (amortization) over its useful life.
totalOperatingExpense	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The totalOperatingExpense represents the expense an entity incurs through its normal business operations such as selling, general and administrative expense, research and development expense, and other operating-related expense
totalOperatingProfit	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The totalOperatingProfit represents the profit or loss derived from it operating activities, before its financial, extraordinary, or other income and charges.

Columns name	Field Type	Standard Values/Format	Definition
totalOperatingProfitPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalOperatingProfit from the previous year of provided financialStatementDate.
financeCosts	Numeric	Unit: Thousands, in the specified currency	The financeCosts represents the expense incurred by the entity while borrowing funds. Usually includes Interest and similar expenses, costs of borrowing, imputed interest elements of financial leases, commitment fees related to obtaining credit facilities and the underwriting of a discount related to long-term provisions.
profitBeforeTaxesAndExtraordinaryExpenses	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The profitBeforeTaxesAndExtraordinaryExpenses represents the profit(loss) result that is related to its ordinary activities before taxation and extraordinary items.
profitBeforeTaxesAndExtraordinaryExpensesPreviousYear	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The profitBeforeTaxesAndExtraordinaryExpenses from the previous year of provided financialStatementDate.
currentIncomeTaxExpense	Numeric	Unit: Thousands, in the specified currency	The currentIncomeTaxExpense represents all income taxes levied on current fiscal year on the income of the entity by governments.
totalExtraordinaryItems	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The totalExtraordinaryItems represents the gain or loss caused by events which are unusual by their nature or which do not occur frequently are defined as Extraordinary income and earnings.
netIncome	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The netIncome represents the profit(loss) reported after expenses and losses have been subtracted from all revenues and gains for the fiscal period, including extraordinary items and discontinued operations.
netIncomePreviousYear	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	The netIncome from the previous year of provided financialStatementDate.
ebitda	Numeric	Unit: Thousands, in the specified currency; Only positive values accepted	EBITDA represents earnings before interest, taxes, depreciation, and amortization.
numberOfEmployees	Numeric		Number of employees for the company.

13.3 Banks Input Template

The same list of minimum fields required that was described for the universal template is also valid for the banks specific template.

Valid Template Upload Information for Banks (Not for Corporates)

Columns name	Field Type	Standard Values/Format	Definition
entityInternationalName	String		The international name for the entity.
entityIdentifier	String		A unique identifier for each entity. This field must be populated.
financialStatementDate	Date	YYYY-MM-DD	The closing date of the financial statement period.

Columns name	Field Type	Standard Values/Format	Definition
asOfDate	Date	YYYY-MM-DD	The date when the data snapshot details are relevant. Currently not relevant for PD calculations.
primaryIndustryClassification	String	NDY, NACE2, NAICS2017	The industry classification for the primary industry. • NDY: Moody's Analytics private companies industry classification • NACE2 as per https://ec.europa.eu/eurostat/web/nace-rev2 • NAICS2017 as per https://www.census.gov/naics/
primaryIndustry	String		The primary industry code for the entity's economic activity. The code must be a valid code of the primaryIndustryClassification. When Orbis provides more than an industry code, the primary code is used.
primaryCountry	String		The primary country for the entity's economic activity. Valid values are the three-letter country codes from ISO (https://www.iso.org/iso-4217-currency-codes.html).
primaryStateProvince	String		The primary state or province for the entity's economic activity. Provide the subdivision ISO 3166 code (https://www.iso.org/iso-3166-country-codes.html) removing the country prefix. For example, the ISO 3166 code for California is US-CA, so simple input CA. Note the field is only currently used for entities in the United States of America.
currency	String		The currency denomination. Valid values are ISO 4217 currency codes. All financial input fields must be expressed in the financial statement currency.
entityLegalForm	String	BranchChamberOffice, C_Corporation, CollectiveEnterprise, Cooperative, Corporation, Farm, ForeignEntity, Government, Institution, JointStockCompany, JointVenture, LimitedLiabilityCompany, LimitedLiabilityPartnership, NonProfit, Other, Partnership, PublicLimitedCompany, S_Corporation, SoleProprietorship, TrustFinancialInstitution, Unknown	The standardized legal forms.
auditQuality	String		The audit opinion provided by an auditor of the financial statement.
totalAssets	Numeric	Unit: Thousands, in the specified currency	totalAssets represents cash, securities, loan and lease receivables, fixed assets, investments in unconsolidated subsidiaries, and other assets.
totalAssetsPreviousYear	Numeric	Unit: Thousands, in the specified currency	The totalAssets from the previous year of provided financialStatementDate.
totalIntangibleAssets	Numeric	Unit: Thousands, in the specified currency	totalIntangibleAssets represents non-physical assets such as mortgage servicing rights,

Columns name	Field Type	Standard Values/Format	Definition
			purchased credit card relationships, all other identifiable intangibles, and goodwill.
shortTermInvestments	Numeric	Unit: Thousands, in the specified currency	shortTermInvestments represents investments with a horizon less than 12 months or one operating cycle.
shortTermNoncoreFunding	Numeric	Unit: Thousands, in the specified currency	shortTermNoncoreFunding represents noncore funding with a remaining maturity less than 12 months or one operating cycle such as federal funds purchased, jumbo CDs, and brokered deposits.
unusedCommitments	Numeric	Unit: Thousands, in the specified currency	unusedCommitments represents the unused portion of the loans that the lender has agreed to be made available to the borrower that the borrower has either not drawn, or has drawn and repaid.
nonperformingAssets	Numeric	Unit: Thousands, in the specified currency	nonperformingAssets represents the credit facilities which the interest and/or installment of principal has remained 'past due' for a specified period, including foreclosed real estate. For US entities, excludes government guaranteed portion of the loans, GNMA loans, and loans covered by FDIC loss-sharing agreements.
debtSecuritiesAndOtherAssets	Numeric	Unit: Thousands, in the specified currency	debtSecuritiesAndOtherAssets represents interest income generating financial instrument asset and other assets excluding real estate owned and other repossessed assets.
otherRealEstateOwned	Numeric	Unit: Thousands, in the specified currency	otherRealEstateOwned represents the book value (not to exceed fair value), less accumulated depreciation, if any, of all real estate other than bank premises owned by the bank and its consolidated subsidiaries.
leaseFinancingReceivables	Numeric	Unit: Thousands, in the specified currency	leaseFinancingReceivables represents outstanding receivable balances relating to direction and leveraged financing leases on property acquired by the bank for leasing purposes.
bankToBankLoans	Numeric	Unit: Thousands, in the specified currency	bankToBankLoans represents the amount of loans to banks and financial institutions.
sovereignLoans	Numeric	Unit: Thousands, in the specified currency	sovereignLoans represents loans to foreign governments and official institutions.
commercialRealEstateLoans	Numeric	Unit: Thousands, in the specified currency	commercialRealEstateLoans represents loans secured by commercial real estate such as nonresidential properties, farmland, multifamily (5 or more) residential properties, or real estate held in foreign offices.
constructionAndDevelopmentLoans	Numeric	Unit: Thousands, in the specified currency	constructionAndDevelopmentLoans represents loans secured by real estate in development and construction stage.
creditCardLoans	Numeric	Unit: Thousands, in the specified currency	creditCardLoans represents credit card-related loans to individuals for personal expenditures.

Columns name	Field Type	Standard Values/Format	Definition
loansToCorporates	Numeric	Unit: Thousands, in the specified currency	loansToCorporates represents the amount of loans to corporates, including small and medium enterprises.
netLoansAndLeases	Numeric	Unit: Thousands, in the specified currency	netLoansAndLeases represents the loans and lease financing receivables, net of unearned income, allowance, and reserve.
otherConsumerLoans	Numeric	Unit: Thousands, in the specified currency	otherConsumerLoans represents amount of loans to consumers excluding credit card loans. Other loans to individuals for household, family, and other personal expenditures (consumer loans) including single payment, installment, student loans, and auto loans.
otherLoans	Numeric	Unit: Thousands, in the specified currency	otherLoans represents other loans including agricultural.
residentialMortgages	Numeric	Unit: Thousands, in the specified currency	residentialMortgages represents loans secured by family residential properties that are usually 1-4 units.
customerDeposits	Numeric	Unit: Thousands, in the specified currency	customerDeposits represents the total customer deposits, excluding deposits from other banks.
totalDeposits	Numeric	Unit: Thousands, in the specified currency	totalDeposits represents the sum of total transaction accounts, non-transaction savings, deposits, and total time deposits.
ifrsDerivativeLiability	Numeric	Unit: Thousands, in the specified currency	ifrsDerivativeLiability represents the IFRS Fair value of derivative financial instruments used for hedging / Portfolio hedge valuation adjustments, recognized as liability. Not necessary for non-IFRS entity.
ifrsDerivativeLiabilityPreviousYear	Numeric	Unit: Thousands, in the specified currency	The ifrsDerivativeLiability from the previous year of provided financialStatementDate.
reservesForLoanLosses	Numeric	Unit: Thousands, in the specified currency	reservesforLoanLosses represents the on balance sheet allowance for loan and lease losses.
netWorth	Numeric	Unit: Thousands, in the specified currency	netWorth represents the sum of all equity items, including retained earnings and other equity. Also known as total shareholder equity.
provisionsForCreditLosses	Numeric	Unit: Thousands, in the specified currency	provisionsForCreditLosses, represents the provisions for loan and lease losses, charges against a bank's operating income which are either likely or certain to be incurred.
netIncome	Numeric	Unit: Thousands, in the specified currency	netIncome represents net interest income plus total noninterest income plus realized gains (losses) on securities and extraordinary items, less total noninterest expense, loan loss provisions, and income taxes
netIncomePreviousYear	Numeric	Unit: Thousands, in the specified currency	The netIncome from the previous year of provided financialStatementDate.

Columns name	Field Type	Standard Values/Format	Definition
modeledUnexpectedLosses	Numeric	Unit: Thousands, in the specified currency	modeledUnexpectedLosses represents the standard deviation of the time series of modeled delinquency rates based on the current portfolio breakdown and historical delinquency rates.
concentrationFactor	Numeric	Unit: Thousands, in the specified currency	concentrationFactor represents Herfindahl index of mean expected loss by asset type.

13.4 NDY - List of values

NDY is the Moody's Analytics companies industry classification. The following table displays the list of values available for both private and listed companies.

Code	Description
N01	AEROSPACE & DEFENSE
N02	AGRICULTURE
N03	AIR TRANSPORTATION
N04	APPAREL & SHOES
N05	AUTOMOTIVE
N06	BANKS AND S&LS
N07	BROADCAST MEDIA
N08	BUSINESS PRODUCTS WHSL
N09	BUSINESS SERVICES
N10	CHEMICALS
N11	COMPUTER HARDWARE
N12	COMPUTER SOFTWARE
N13	CONSTRUCTION
N14	CONSTRUCTION MATERIALS
N15	CONSUMER DURABLES
N16	CONSUMER DURABLES RETL/WHSL
N17	CONSUMER PRODUCTS
N18	CONSUMER PRODUCTS RETL/WHSL
N19	CONSUMER SERVICES
N20	ELECTRICAL EQUIPMENT
N21	ELECTRONIC EQUIPMENT

Code	Description
N22	ENTERTAINMENT & LEISURE
N23	FINANCE COMPANIES
N24	FINANCE NEC
N25	FOOD & BEVERAGE
N26	FOOD & BEVERAGE RETL/WHSL
N27	FURNITURE & APPLIANCES
N28	HOTELS & RESTAURANTS
N29	INSURANCE - LIFE
N30	INSURANCE - PROP/CAS/HEALTH
N31	INVESTMENT MANAGEMENT
N32	LESSORS
N33	LUMBER & FORESTRY
N34	MACHINERY & EQUIPMENT
N35	MEASURE & TEST EQUIPMENT
N36	MEDICAL EQUIPMENT
N37	MEDICAL SERVICES
N38	MINING
N39	OIL REFINING
N40	OIL, GAS & COAL EXPL/PROD
N41	PAPER
N42	PHARMACEUTICALS
N43	PLASTIC & RUBBER
N44	PRINTING
N45	PUBLISHING
N46	REAL ESTATE
N47	REAL ESTATE INVESTMENT TRUSTS
N48	SECURITY BROKERS & DEALERS
N49	SEMICONDUCTORS
N50	STEEL & METAL PRODUCTS
N51	TELEPHONE
N52	TEXTILES

Code	Description
N53	TOBACCO
N54	TRANSPORTATION EQUIPMENT
N55	TRANSPORTATION
N56	TRUCKING
N57	UNASSIGNED
N58	UTILITIES NEC
N59	UTILITIES, ELECTRIC
N60	UTILITIES, GAS
N61	CABLE TV

14 Appendix: Output Details

The API response depends on the endpoint used. This appendix is for two main categories of output:

- PD analytics
- Credit Limits analytics

14.1 PD Analytics

The PD analytics in the response can be grouped into two categories:

- Standard outputs: the set of outputs returned to the user, irrespective of the entity characteristics.
- Entity and Country Specific outputs: differ depending on the characteristics of the entities in analysis.

These outputs can be accessed using the endpoints:

- POST- /edfx/v1/entities/pds
- POST- /edfx/v1/entities/pds/riskcalc
- POST- /edfx/v1/entities/modelInputs to upload data via CSV file (for more details check the relevant paragraph in [Input Fields](#) section), followed by a call to one of the PD endpoints above to obtain the outputs.
- POST -/edfx/v1/entities/payment

The output is in JSON format. When the response volume is big, a pre-signed URL is generated; in this case the output is a [CSV](#) file that includes all input columns as well as additional columns related to the results.

All input columns are described in the [Input Fields](#) section. All the results columns are described in the following section.

Note Values related to the inputs fields correspond to imported values after some transformations, like a currency conversion. Import only Company information (no Financial Statements information) for some entities, the process imputes a value to the main input columns left empty.

When the request uses the Advanced Option `asyncResponse = true`, the PD is accessed using the batch endpoint: GET - /edfx/v1/processes/{processId}/files

In this case, the API returns a JSON file with the same characteristics and fields described in the following paragraphs.

14.1.1 Standard Outputs

The list of standard outputs is presented in the following table. The set of output fields that are returned, irrespective of the entity characteristics and Advanced Options.

Field	Type and Format	Definition	Example
entityId	String	The BVD ID of a private entity or the PID of the listed company.	
asOfDate	Date YYYY-MM-DD	The date when calculations are done. This date is the "partition" of data used to calculate the output.	2020-01-01
pd	Number	One-year Probability of Default estimate	0.0024
impliedRating	String	The rating implied from the calculated PD. Mapping is based on a static scale for private entities and on a dynamic scale for listed companies.	AA
legalForm	String	The standardized legal forms. The possible values are the same described in the Input Details page for entityLegalForm. This is only relevant for unlisted companies.	LimitedLiabilityCompany
legalStatus	String	The status of the entity. The possible values are the same described in the Input Details page for entityStatus. This is only relevant for unlisted companies.	Active
confidence	String	Confidence is an indicator that provides concise information about elements playing a role in the calculations. It is available in two formats, the code and the description. The indicator is a combination of the following flags: - Depending on the entity's legal status in Orbis: - BKRPT is returned when legal status is Bankruptcy. - ACT_IMP is returned when legal status is Active but impaired (default, insolvency, or rescue plans). - NOT_ACT is returned when legal status indicates liquidation, inactivity, or a dissolved entity. - If none of these flags are part of the indicator, the entity's legal status is Active (including reorganization and dormant cases), unknown, or not assigned. This flag is only assigned for entities that are present in Orbis, not for entities uploaded by the user. - Depending on the entity type and the amount of information available for the company under analysis: - P indicates a Public Company (these are listed companies). - PF indicates a Private company with full financials. - PP indicates a Private company with limited financials. - PN indicates a Private company with no financials. - PY indicates a Private company with no financials but payment data available. - Depending on the quality of the input financial statement: - G indicates good quality input data.	PF-G-R ; Private firm model with full financials available, recent, and good quality inputs. PN-X-X-FLAG_T-NR ; Private firm model with no financials available. Model not recommended for entity with this entity type. Industry not available, but proxy is assigned.

Field	Type and Format	Definition	Example
		- I indicates that the input data failed validation Business Rules. 4. Depending on how recent the financial statement is: - R indicates recent financials. Statement date is maximum 18 months old. - O indicated old financial statements. Between 18 and 36 months (included) old. - S indicates stale financial statements, Older than 36 months. 5. Depending on the entity's Legal Form, Entity Type, and PID (for listed entities only): - FLAG_L indicates that the model used for PD calculation is not recommended for this legal entity. FLAG_L is assigned when the entity is a Nonprofit organization or a Public limited company whose PID is not available. - FLAG_T indicates the model used for PD calculation is not recommended for an entity with this entity type. FLAG_T is assigned when the entity is a Public authority, State, Government, a Foundation, Research institute, or Marine vessel. 6. Depending on the entity's industry classification code: - NN is assigned to indicate that no industry code is available. - NR is assigned when there is no industry code available, but the scoring process could assign a proxy. This happens when the entity is a Bank, Mutual and pension fund, Nominee, Trust, Trustee, Financial company, Insurance company, Private equity firm, Venture capital, Hedge fund, Marine vessel, Foundation, Research institute whose PD is calculated using the Benchmark model. "X" is added in the code as a placeholder if part of the information is not relevant.	
termStructure		The PD term structure, produced over a 10 years horizon. Three types of term structure are available in the outputs: - Annualized - Cumulative - Forward The related implied rating is also available, over 10 years horizon.	
annualized		The annualized value represents the cumulative value for a given period, stated on a per-year basis. For example, a company with a cumulative 5-year PD value of 13.44% would have a 5-year annualized PD value of 2.84% obtained as per the following formula. This means that the average default rate per year for a 13.44% cumulative default rate is 2.84%. $Annualized\ 5y\ PD = 1 - (1 - Cumulative\ 5y\ PD)^{1/5}$ For each year of term structure, the API returns the PD values with labels "annualized1y, annualized2y, ..."	

Field	Type and Format	Definition	Example
annualize dTy	String	Probability of default at year t of the term structure (with t=1...10)	0.0027
cumulative		The cumulative PD term structure is available. This represents the probability that a firm defaults between years 0 and T, where T represents a year along the term structure. For example, a five-year cumulative PD value represents the likelihood that a firm defaults between years 0 and 5. For the cumulative PD term structure, the API returns the PD values with labels "cumulative1y, cumulative2y, ..."	
forward		The forward PD terms structure is available. This represents the probability that a firm defaults between years T-1 and T (where T represents a year along the term structure), conditional upon survival until year T-1. For example, a three-year forward PD value represents the likelihood that a firm defaults between years 2 and 3 based on the condition that the firm remains operational until year 2. $Forward\ 5y\ PD = 1 - (1 - Cumulative\ 5y\ PD) / (1 - Cumulative\ 4y\ PD)$ For the cumulative PD term structure, the API returns the PD values with labels "forward1y, forward2y, ..."	
impliedRating		The implied rating consistent with the risk level of the displayed PDs, over the same 10 years horizon. The API returns the PD values with labels "impliedRating1y, impliedRating2y, ..."	
History		All the metrics explained previously are also provided for the historic time series. In the "history" section of the API response, these metrics are repeated for each calculation date (asOfDate) in the time series. The calculation dates are determined considering the length of the time series determined by the startDate and endDate provided in the API request.	

14.1.2 Entity and Country-Specific Outputs

Additional details are available and accessible to the API user. This group of fields is different depending on whether the entity is unlisted or listed. Furthermore, for unlisted entities the list of specific outputs is also variable depending on the entity's country. The expanded list of output fields can be generated by setting up some Advanced Options in the API request (See [Input Fields](#) page for further description). When you enable Advanced Options, the following information is accessible.

Group API outputs in three categories:

- Related to calculations methodology
- Related to calculations input details
- Related to calculations results details

Related to calculations methodology

This part of the API response gives more details regarding the model involved in the PD calculations and how to access the relevant documentation available.

Field	Field Type and Format	Definition	Example
modelDetails		Information related to the model involved in the calculation of the PD. These details are only available when modelDetail=true in the API request.	
modelId	String	ID of the model applied. See Appendix: Models Inventory for the list of available models.	GBR40
documentation	String	documentName and link (URL) to download the documentation related to the model applied.	

Related to calculations input details

This part of the API response is dedicated to listing inputs used in the PD calculations. They differ depending if the companies are private or listed. Furthermore, the content is different for private companies that receive a benchmark PD or a payment PD.

Private companies

For Private companies, this list has two parts: a fixed part that includes fields always served in this section, and a variable part that includes Country specific fields, this changes depending on the company under analysis. The fields always served in this section of the API response are described in the following table.

Field	Field Type and Format	Definition	Example
inputData		Information related to the inputs used as drivers in the PD calculation. These details are only available when inputDetail=true in the API request.	
Currency	String	The currency used in the PD calculation methodology.	
Current date - month	Number	The month and year of the asOfDate.	"current date - month": "",
Current date - year	Number		"current date - year": "",
Statement date - month	Number	The month and year of the financial statement date.	"statement date - month": "12",
Statement date - year	Number		"statement date - year": "2020",

Field	Field Type and Format	Definition	Example
Financial statement only EDF	String	EDF calculation mode.	"financial statement only EDF": "on",
Industry classification Industry definition	String String	NDY classification is shown in this field. If the user provided a code from a different industry classification (via input file upload), the input is mapped to NDY during the calculation.	"industry classification": "NDY", "industry definition": "N16",
User-defined time period		Not relevant.	"user-defined time period": ""

For the previously described fields, some Country-specific fields are also served in the inputData section. Depending on the type of entity under analysis and on its primary Country, the PD calculations require the application of different Country Adjustment Factors and the calculation of different financial ratios. As a result, the list of input fields needed for the PD calculations change.

For example, the PD calculations for a corporate in the United Kingdom needs inputs, like *Provisions for liabilities and charges*, that are not needed to calculate the PD of a Belgian company.

For listed companies, the specific items displayed in this section are related to:

- Financial Statements - This section displays information about current and long-term liabilities and debt, minority interests, deferred taxes, common, and preferred dividends.
- Market Variables - Information related to asset volatility, market value of asset, share price, and other market variables that play a key role in the PD calculation.

Benchmark PD

For private companies that receive a benchmark PD due to lack of financial statements, the information in inputData includes

Field	Field Type and Format	Definition	Example
inputData			
ndy	String	The industry classification for the primary industry in NDY i.e. Moody's Analytics private companies industry classification.	N09
country	String	The primary country for the entity's economic activity in three-letter ISO code format.	GBR
state	String	The primary state or province for the entity's economic activity in subdivision ISO 3166 code removing the country prefix. Relevant for United States of America only.	

Field		Field Type and Format	Definition	Example
	size	String	Imputed size. The imputation is done as part of the Size and Leverage Imputation engine, starting from the company's total asset, number of employees or turnover as well as country and industry information. Possible values: ▪ Extra Small (<100,000) ▪ Small [100,000-3,000,000) ▪ Medium [3,000,000-15,000,000) ▪ Large [15,000,000-150,000,000) ▪ Extra Large [150,000,000+)	Small
	leverage	String	Imputed leverage bucket. The imputation is done as part of the Size and Leverage Imputation engine, starting from the company's total asset, and total liabilities (where these values are available) as well as country and industry information. Possible values: ▪ Low (0-30] ▪ Medium (30-70] ▪ High (70-100] ▪ Over-leveraged (>100)	High

Payment PD

For companies in the USA that have no financials information but have payment data available, the user can access payment PDs; the Payment model uses the following inputs.

Field		Field Type and Format	Definition	Example
inputData				
	incorporationYear	Date		
	30dpdTradelineGrowth6m	Number	Growth in the percentage of tradelines that are 30 or more days past due.	0.05714286
	tradelineSpendVolatility6m	Number	Volatility of spending in the past 6 months.	0.22852133
	90dpdTradelinePercentage12m	Number	Percentage of the firm's providers with tradelines that are more than 90 days past due.	0.03352545
	tradelinePaymentsAverage12m	Number	Average percentage of tradelines paid in the last 12 months.	0.58212593
	tradelinePaymentsVolatility6m	Number	Volatility of payments in the past 6 months.	0.09256662
	operationsTradelinePayments	Number	Average percentage of tradelines classified as "Operations" paid in the last 12 months.	0.58212593

Field	Field Type and Format	Definition	Example
tsAverage12m			

Related to calculations results details

This section of the API response is dedicated to listing the output produced by the PD calculations, and key elements useful in analyzing outputs. For Private companies, the fields displayed in this section can be grouped in the categories described in the following table. Since the PD calculations results are different depending on the entity's country and characteristics, the actual list of fields in these categories is variable.

Field	Field Type and Format	Definition	Example
resultData		Information related to the PD calculations results, and to the key elements that are useful in analyzing those results. All the inputs in the inputData rows described in the preceding section are also included in this part (except for statement date month and year). These details are only available when <code>resultDetail=true</code> in the API request.	
EDF Term Structure	String	The PD term structure in the three formats also available in the main results part of the output: annualized, cumulative, and forward (as described in the preceding section). The values related to the annualized term structure are prefixed with "annualized T-yr. EDF: "; the values related to the cumulative term structure are prefixed with "cumulative T-yr. EDF: "; the values related to the forward term structure are prefixed with "forward T-yr. EDF: ".	"Annualized 5-Yr. EDF": "0.01801634916", "Cumulative 5-Yr. EDF": "0.08689381158", "Forward 5-Yr. EDF": "0.01827322675"
Percentile	String	The percentile measure compares a firm's one-year and five-year PD values to the applicable PD values for all firms in the model development sample. The percentile reflects all firms in the model development sample that exhibit a PD value lower than the firm under analysis.	"one-year percentile": "9.45%", "five-year percentile": "7.20%"
Bond Default Rate Mapping	String	The PD Implied Rating.	"Bond Default Rate Mapping (1 Yr)": "Ba2.edf", "Bond Default Rate Mapping (5 Yr)": "Ba3.edf"
Financial Ratios	Number	The model specific list of the financial ratios that drive PD. The ratio with the highest value is the highest risk contributor. All these ratios are prefixed with "ratio:" and they are presented as raw numeric value of the ratio or level calculated.	"ratio: real total assets": "8299781591.82"

Field	Field Type and Format	Definition	Example
Financial Ratios Relative Contributions	String	Quantifies how much each ratio contributes to the 1-year and 5-year PD values, respectively. A positive contribution value means the ratio increases the firm's PD value and its risk, relative to the average default rate observed in a country. A negative contribution value means that the ratio decreases the firm's PD value and its risk, relative to the average default rate observed in a country. All these ratios are prefixed with "rel. contrib. 1 yr:" or "rel. contrib. 5 yr:" and they are presented as percentages wrapped in a string format.	"rel. contrib. 1 yr: real total assets": "-23.22%", "rel. contrib. 5 yr: real total assets": "-34.73%"
Financial Ratios Relative Sensitivity	String	Quantifies how a change in a ratio affects the 1-year and 5-year PD values, respectively. A positive sensitivity value means the change in a firm's PD value is in the same direction as the change in the ratio. A negative sensitivity value means the change in a firm's PD value is in the opposite direction of the change in the ratio. All these ratios are prefixed with "rel. sens. 1 yr:" or "rel. sens. 5 yr:" and they are presented as percentages wrapped in a string format.	"rel. sens. 1 yr: real total assets": "-5.42%", "rel. sens. 5 yr: real total assets": "-8.96%"
Financial Ratios Percentiles	String	This is the result of the comparison of a firm's PD value to that of all firms in the model development sample. It indicates the percentage of firms in the sample that are less risky and have PD values lower than the PD value of the firm under analysis. All these percentiles are prefixed with "%-tile:" and they are presented as percentages wrapped in a string format.	"%-tile: real total assets": "98.88%"

Payment PD

Also for US companies that receive a payment PD, the results details will show the relative contribution of each one of the input variable explained above as well as the variables' related percentiles.

Public Firms

For listed companies, the correspondent information is already fully described in the inputData section, hence it is not repeated here.

14.2 Credit Limits

The results related to the Credit Limits can be accessed through the endpoint dedicated to this tool:

POST - /edfx/v1/tools/creditLimit

The output fields available for Credit Limits is described in the following table.

Field	Field Type and Format	Definition	Example
entityId	String	The BVD ID of a private entity or the PID of the listed company.	
creditLimits			
asOfDate	Date	The date when calculations are done. This date is the "partition" of data used to calculate the output.	
pd	Numeric	One-year Probability of Default	
balanced	Numeric	The value of the credit limit corresponding to balanced risk taking assumptions	
aggressive	Numeric	The value of the credit limit corresponding to aggressive risk taking assumptions	
conservative	Numeric	The value of the credit limit corresponding to conservative risk taking assumptions	
currency	String	The currency in which the credit limit is expressed. The output of the credit limits calculations is always expressed in USD.	USD
unit	Numeric	This is the unit in which the credit limit is expressed; it is expressed in the currency mentioned in the currency field. For credit limits, this is always one and it stands for one USD.	1

15 Appendix: Models Inventory

The table below provides a mapping between model ID and the model name for the underlying models used to calculate the Moody's Analytics PD.

Model ID	Model Name	Type of Model
AUS40	RiskCalc 4.0 Australia	Private Firms
AUT40	RiskCalc 4.0 Austria	Private Firms
BEL40	RiskCalc 4.0 Belgium	Private Firms
BNK40	RiskCalc 4.0 Banks	Private Firms
CAN32	RiskCalc 3.2 Canada	Private Firms
EDF9	CreditEdge EDF 9.0	Public Firms
CHE31	RiskCalc 3.1 Switzerland	Private Firms
CHN40	RiskCalc 4.0 China	Private Firms
DEU40	RiskCalc 4.0 Germany	Private Firms
DNK40	RiskCalc 4.0 Denmark	Private Firms
EMM31	RiskCalc 3.1 Emerging Markets	Private Firms
ESP40	RiskCalc 4.0 Spain	Private Firms
FIN40	RiskCalc 4.0 Finland	Private Firms
FRA40	RiskCalc 4.0 France	Private Firms
GBR40	RiskCalc 4.0 United Kingdom	Private Firms
GRC40	RiskCalc 4.0 Greece	Private Firms
HMARK/BEN10	Benchmark 1.0	Private Firms with incomplete information, with no statements, with old statements.
ITA40	RiskCalc 4.0 Italy	Private Firms
JPN32	RiskCalc 3.2 Japan	Private Firms
KOR31	RiskCalc 3.1 Korea	Private Firms
LFE40	RiskCalc 4.0 Europe Large Firm	Private Firms

Model ID	Model Name	Type of Model
LNA31	RiskCalc 3.1 North America Large Firm	Private Firms
NLD40	RiskCalc 4.0 Netherland	Private Firms
NOR40	RiskCalc 4.0 Norway	Private Firms
PAY10	Payment 1.0	Small/Medium private firms in USA with no financials but payment data available.
PRT40	RiskCalc 4.0 Portugal	Private Firms
RUS31	RiskCalc 3.1 Russia	Private Firms
SGP40	RiskCalc 4.0 Singapore	Private Firms
SWE40	RiskCalc 4.0 Sweden	Private Firms
USA40	RiskCalc 4.0 United States Corporates	Private Firms
USF40	RiskCalc 4.0 United States Banks	Private Firms
ZAF32	RiskCalc 3.2 South Africa	Private Firms

16 Appendix: Peer Groups - Country to Region Mapping

List of each country ISO code and the related region associated for the purpose of peer groups aggregation.

Country	Country ISO Code	Peer Region
Aruba	ABW	SOAM
Afghanistan	AFG	MIDENOAF
Angola	AGO	CESOAM
Anguilla	AIA	SOAM
Åland Islands	ALA	WESTEUR
Albania	ALB	EEURRUS
Andorra	AND	WESTEUR
United Arab Emirates	ARE	MIDENOAF
Argentina	ARG	SOAM
Armenia	ARM	EEURRUS
American Samoa	ASM	AUSNZ
Antarctica	ATA	SOAM
French Southern Territories	ATF	AUSNZ
Antigua and Barbuda	ATG	SOAM
Australia	AUS	AUSNZ
Austria	AUT	WESTEUR
Azerbaijan	AZE	EEURRUS
Burundi	BDI	MIDENOAF
Belgium	BEL	WESTEUR
Benin	BEN	CESOAM
Bonaire, Sint Eustatius and Saba	BES	USCAN
Burkina Faso	BFA	CESOAM

Country	Country ISO Code	Peer Region
Bangladesh	BGD	SOASIA
Bulgaria	BGR	EEURRUS
Bahrain	BHR	CESOAM
Bahamas	BHS	USCAN
Bosnia and Herzegovina	BIH	EEURRUS
Saint Barthélemy	BLM	SOAM
Belarus	BLR	EEURRUS
Belize	BLZ	SOAM
Bermuda	BMU	USCAN
Bolivia	BOL	SOAM
Brazil	BRA	SOAM
Barbados	BRB	SOAM
Brunei Darussalam	BRN	SEASIA
Bhutan	BTN	SOASIA
Bouvet Island	BVT	CESOAM
Botswana	BWA	CESOAM
Central African Republic	CAF	CESOAM
Canada	CAN	USCAN
Cocos (Keeling) Islands	CCK	AUSNZ
Switzerland	CHE	WESTEUR
Chile	CHL	SOAM
China	CHN	NOASIA
Côte d'Ivoire	CIV	MIDENOAF
Cameroon	CMR	CESOAM
Congo (the Democratic Republic of the)	COD	CESOAM
Congo	COG	CESOAM
Cook Islands	COK	AUSNZ
Colombia	COL	SOAM
Comoros	COM	MIDENOAF
Cabo Verde	CPV	CESOAM
Costa Rica	CRI	SOAM

Country	Country ISO Code	Peer Region
Cuba	CUB	SOAM
Curaçao	CUW	SOAM
Christmas Island	CXR	SEASIA
Cayman Islands	CYM	USCAN
Cyprus	CYP	WESTEUR
Czechia	CZE	EEURRUS
Germany	DEU	WESTEUR
Djibouti	DJI	MIDENOAF
Dominica	DMA	SOAM
Denmark	DNK	WESTEUR
Dominican Republic	DOM	SOAM
Algeria	DZA	MIDENOAF
Ecuador	ECU	SOAM
Egypt	EGY	MIDENOAF
Eritrea	ERI	MIDENOAF
Western Sahara	ESH	CESOAM
Spain	ESP	WESTEUR
Estonia	EST	EEURRUS
Ethiopia	ETH	MIDENOAF
Finland	FIN	WESTEUR
Fiji	FJI	AUSNZ
Falkland Islands	FLK	WESTEUR
France	FRA	WESTEUR
Faroe Islands	FRO	WESTEUR
Micronesia	FSM	AUSNZ
Gabon	GAB	CESOAM
United Kingdom of Great Britain and Northern Ireland	GBR	WESTEUR
Georgia	GEO	EEURRUS
Guernsey	GGY	WESTEUR
Ghana	GHA	CESOAM
Gibraltar	GIB	WESTEUR

Country	Country ISO Code	Peer Region
Guinea	GIN	CESOAM
Guadeloupe	GLP	SOAM
Gambia	GMB	CESOAM
Guinea-Bissau	GNB	CESOAM
Equatorial Guinea	GNQ	CESOAM
Greece	GRC	WESTEUR
Grenada	GRD	SOAM
Greenland	GRL	WESTEUR
Guatemala	GTM	SOAM
French Guiana	GUF	SOAM
Guam	GUM	AUSNZ
Guyana	GUY	SOAM
Hong Kong	HKG	NOASIA
Heard Island and McDonald Islands	HMD	AUSNZ
Honduras	HND	SOAM
Croatia	HRV	EEURRUS
Haiti	HTI	SOAM
Hungary	HUN	EEURRUS
Indonesia	IDN	SEASIA
Isle of Man	IMN	WESTEUR
India	IND	SOASIA
British Indian Ocean Territory	IOT	SOASIA
Ireland	IRL	WESTEUR
Iran	IRN	MIDENOAF
Iraq	IRQ	MIDENOAF
Iceland	ISL	WESTEUR
Israel	ISR	MIDENOAF
Italy	ITA	WESTEUR
Jamaica	JAM	SOAM
Jersey	JEY	WESTEUR
Jordan	JOR	MIDENOAF

Country	Country ISO Code	Peer Region
Japan	JPN	NOASIA
Kazakhstan	KAZ	EEURRUS
Kenya	KEN	MIDENOAF
Kyrgyzstan	KGZ	EEURRUS
Cambodia	KHM	SEASIA
Kiribati	KIR	AUSNZ
Saint Kitts and Nevis	KNA	SOAM
Korea (the Republic of)	KOR	NOASIA
Kuwait	KWT	MIDENOAF
Lao People's Democratic Republic	LAO	SEASIA
Lebanon	LBN	MIDENOAF
Liberia	LBR	CESOAM
Libya	LBY	MIDENOAF
Saint Lucia	LCA	SOAM
Liechtenstein	LIE	WESTEUR
Sri Lanka	LKA	SOASIA
Lesotho	LSO	CESOAM
Lithuania	LTU	EEURRUS
Luxembourg	LUX	WESTEUR
Latvia	LVA	EEURRUS
Macao	MAC	NOASIA
Saint Martin	MAF	SOAM
Morocco	MAR	MIDENOAF
Monaco	MCO	WESTEUR
Moldova	MDA	EEURRUS
Madagascar	MDG	CESOAM
Maldives	MDV	SOASIA
Mexico	MEX	SOAM
Marshall Islands	MHL	USCAN
Republic of North Macedonia	MKD	EEURRUS
Mali	MLI	CESOAM

Country	Country ISO Code	Peer Region
Malta	MLT	WESTEUR
Myanmar	MMR	SEASIA
Montenegro	MNE	EEURRUS
Mongolia	MNG	NOASIA
Northern Mariana Islands	MNP	AUSNZ
Mozambique	MOZ	CESOAM
Mauritania	MRT	EEURRUS
Montserrat	MSR	SOAM
Martinique	MTQ	SOAM
Mauritius	MUS	CESOAM
Malawi	MWI	MIDENOAF
Malaysia	MYS	SEASIA
Mayotte	MYT	MIDENOAF
Namibia	NAM	CESOAM
New Caledonia	NCL	AUSNZ
Niger	NER	CESOAM
Norfolk Island	NFK	WESTEUR
Nigeria	NGA	CESOAM
Nicaragua	NIC	SOAM
Niue	NIU	AUSNZ
Netherlands	NLD	WESTEUR
Norway	NOR	WESTEUR
Nepal	NPL	SEASIA
Nauru	NRU	AUSNZ
New Zealand	NZL	AUSNZ
Oman	OMN	MIDENOAF
Pakistan	PAK	SOASIA
Panama	PAN	SOAM
Pitcairn	PCN	AUSNZ
Peru	PER	SOAM
Philippines	PHL	SEASIA

Country	Country ISO Code	Peer Region
Palau	PLW	SEASIA
Papua New Guinea	PNG	AUSNZ
Poland	POL	EEURRUS
Puerto Rico	PRI	SOAM
Korea (the Democratic People's Republic of)	PRK	NOASIA
Portugal	PRT	WESTEUR
Paraguay	PRY	SOAM
Palestine, State of	PSE	MIDENOAF
French Polynesia	PYF	AUSNZ
Qatar	QAT	MIDENOAF
Réunion	REU	AUSNZ
Romania	ROU	EEURRUS
Russian Federation	RUS	EEURRUS
Rwanda	RWA	MIDENOAF
Saudi Arabia	SAU	MIDENOAF
Sudan	SDN	MIDENOAF
Senegal	SEN	CESOAM
Singapore	SGP	SEASIA
South Georgia and the South Sandwich Islands	SGS	SOAM
Saint Helena, Ascension and Tristan da Cunha	SHN	CESOAM
Svalbard and Jan Mayen	SJM	WESTEUR
Solomon Islands	SLB	AUSNZ
Sierra Leone	SLE	CESOAM
El Salvador	SLV	SOAM
San Marino	SMR	WESTEUR
Somalia	SOM	CESOAM
Saint Pierre and Miquelon	SPM	USCAN
Serbia	SRB	EEURRUS
South Sudan	SSD	CESOAM
Sao Tome and Principe	STP	CESOAM
Suriname	SUR	SOAM

Country	Country ISO Code	Peer Region
Slovakia	SVK	EEURRUS
Slovenia	SVN	EEURRUS
Sweden	SWE	WESTEUR
Eswatini	SWZ	CESOAM
Sint Maarten	SXM	USCAN
Seychelles	SYC	CESOAM
Syrian Arab Republic	SYR	MIDENOAF
Turks and Caicos Islands	TCA	SOAM
Chad	TCD	CESOAM
Togo	TGO	CESOAM
Thailand	THA	SEASIA
Tajikistan	TJK	EEURRUS
Tokelau	TKL	AUSNZ
Turkmenistan	TKM	EEURRUS
Timor-Leste	TLS	SEASIA
Tonga	TON	AUSNZ
Trinidad and Tobago	TTO	SOAM
Tunisia	TUN	MIDENOAF
Turkey	TUR	MIDENOAF
Tuvalu	TUV	AUSNZ
Taiwan (Province of China)	TWN	NOASIA
Tanzania, United Republic of	TZA	CESOAM
Uganda	UGA	CESOAM
Ukraine	UKR	EEURRUS
United States Minor Outlying Islands	UMI	AUSNZ
Uruguay	URY	EEURRUS
United States of America	USA	USCAN
Uzbekistan	UZB	EEURRUS
Holy See	VAT	WESTEUR
Saint Vincent and the Grenadines	VCT	SOAM
Venezuela (Bolivarian Republic of)	VEN	SOAM

Country	Country ISO Code	Peer Region
Virgin Islands (British)	VGB	USCAN
Virgin Islands (U.S.)	VIR	USCAN
Viet Nam	VNM	SEASIA
Vanuatu	VUT	AUSNZ
Wallis and Futuna	WLF	AUSNZ
Samoa	WSM	AUSNZ
Yemen	YEM	MIDENOAF
South Africa	ZAF	CESOAM
Zambia	ZMB	CESOAM
Zimbabwe	ZWE	CESOAM

17 Appendix: Supported Identifiers

List of identifier supported in the relevant search endpoints.

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
isin	World	ISIN number	Securities number	DE0007500001
ticker	World	Ticker symbol	Stock number	TKA
entityIdentifierCik	World	CIK	Company ID number	0000104169
entityIdentifierCuNcuaNumber	World	CU/NCUA number	Company ID number	5536
entityIdentifierEscbInvestmentFundId	World	ESCB Investment fund ID	Company ID number	AT0000032489922
entityIdentifierFdicCertificateNumber	World	FDIC Certificate Number	Company ID number	61550
entityIdentifierFdicUniqueNumber	World	FDIC Unique Number	Company ID number	40088
entityIdentifierImoNumber	World	IMO number	Company ID number	1009132
lei	World	LEI	Company ID number	253400N2R0RF14JQ0060
entityIdentifierRssdId	World	RSSD ID	Company ID number	2654818
entityIdentifierSwiftCode	World	SWIFT code	Company ID number	NRWBDEDM
	World	TIN	Company ID number	40003468507
figi	World	Financial Instrument Global Identifier (FIGI)	Securities number	BBG000BLNNV0
entityIdentifierBvd	World	BvD ID	Securities number	US911144442
pid	World	Moody's Identifier (PID)	Securities number	34537A
entityIdentifierNipt	Albania	NIPT	VAT/Tax number	K72410014H
entityIdentifierNrcNumber	Algeria	NRC number	Trade register number	00B0013767
entityIdentifierNis	Algeria	NIS	Company ID number	099806010706436
entityIdentifierNif	Angola	NIF	VAT/Tax number	5402133301
entityIdentifierCuit	Argentina	CUIT	VAT/Tax number	30-54668997-9

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierNin	Armenia	NIN	Trade register number	269110815193
entityIdentifierTaxNumber	Armenia	Tax number	VAT/Tax number	02259659
entityIdentifierAbn	Australia	ABN	Trade register number	51 064 874 531
entityIdentifierAcn	Australia	ACN	Company ID number	064 874 531
entityIdentifierRbn	Australia	RBN	Trade register number	T3622608-NSW
entityIdentifierOenbNumber	Austria	OeNB number	Company ID number	5921155
entityIdentifierTaxNumber	Austria	Tax number	VAT/Tax number	079/8331
entityIdentifierTradeRegister Number	Austria	Trade register number	Trade register number	FN 172048 m (Wien)
entityIdentifierUstIdnr	Austria	UST-IdNr.	VAT/Tax number	ATU51507409
entityIdentifierTaxNumber	Azerbaijan	Tax number	VAT/Tax number	1700382471
entityIdentifierTradeRegister Number	Bahamas	Trade register number	Trade register number	68891B
entityIdentifierCrNumber	Bahrain	CR number	Trade register number	8919
entityIdentifierTradeRegister Number	Bangladesh	Trade register number	Trade register number	C-62833
entityIdentifierTradeRegister Number	Barbados	Trade register number	Trade register number	34142C
entityIdentifierInn	Belarus	INN	VAT/Tax number	100219778
entityIdentifierOkpo	Belarus	OKPO	Statistical number	00154387
entityIdentifierBelgianCompanyNumber	Belgium	Belgian company number	Trade register number	0417.497.106
entityIdentifierEstablishment Number	Belgium	Establishment number	Company ID number	2.143.775.125
entityIdentifierNssoCode	Belgium	NSSO code	Company ID number	1548572-01
entityIdentifierTradeRegister Number	Belize	Trade register number	Trade register number	100094
entityIdentifierRccmNumber	Benin	RCCM number	Trade register number	RB/COT/08B2813
entityIdentifierTradeRegister Number	Bermuda	Trade register number	Trade register number	19316
entityIdentifierMatriculaDeComercio	Bolivia	Matricula de Comercio	Trade register number	00013025
entityIdentifierNit	Bolivia	NIT	VAT/Tax number	00013058
entityIdentifierJib	Bosnia and Herzegovina	JIB	VAT/Tax number	4200225150005
entityIdentifierUcn	Bosnia and Herzegovina	UCN	Trade register number	11033156
entityIdentifierCnpj	Brazil	CNPJ	VAT/Tax number	33.000.167/0001-01

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierNire	Brazil	NIRE	Trade register number	35300330587
entityIdentifierPin	Bulgaria	PIN	Trade register number	4011195302
entityIdentifierUic	Bulgaria	UIC	Trade register number	812114069
	Cambodia	TIN	VAT/Tax number	K005-105019003
entityIdentifierTradeRegister Number	Cambodia	Trade register number	Trade register number	00003084
entityIdentifierNiu	Cameroon	NIU	VAT/Tax number	M024800000316X
cusip	Canada	CUSIP number	Securities number	345370
entityIdentifierBusinessNumber	Canada	Business Number	VAT/Tax number	102185832
entityIdentifierCorporationNumber	Canada	Corporation Number	Company ID number	007478-1
entityIdentifierRut	Chile	RUT	VAT/Tax number	90.690.000-9
entityIdentifierNoc	China	NOC	Statistical number	710926094
entityIdentifierTaxNumber	China	Tax number	VAT/Tax number	110105710926094
entityIdentifierTradeRegister Number	China	Trade register number	Trade register number	100000000032985
entityIdentifierUnifiedSocial CreditCode	China	Unified social credit code	Trade register number	91110000710926094P
entityIdentifierNit	Colombia	NIT	VAT/Tax number	8001972684
entityIdentifierTradeRegister Number	Costa Rica	Trade register number	Trade register number	4-000-042139
entityIdentifierBrojObrtnice	Croatia	Broj obrtnice	Trade register number	92963223473
entityIdentifierMbo	Croatia	MBO	Statistical number	92040101
entityIdentifierMbs	Croatia	MBS	Trade register number	080020970
entityIdentifierNin	Croatia	NIN	Statistical number	03449602
entityIdentifierOib	Croatia	OIB	VAT/Tax number	05937759187
entityIdentifierTradeRegister Number	Cyprus	Trade register number	Trade register number	C337518
	Cyprus	VAT number	VAT/Tax number	CY10145530D
entityIdentifierTradeRegister Number	Czech Republic	Trade register number	Trade register number	00177041
	Czech Republic	VAT number	VAT/Tax number	CZ00177041
entityIdentifierCvrNummer	Denmark	CVR-nummer	Trade register number	22756214
entityIdentifierPNummer	Denmark	P-nummer	Company ID number	1001550956

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierTradeRegisterNumber	Denmark (Faroe Islands)	Trade register number	Trade register number	1724
entityIdentifierGerNumber	Denmark (Greenland)	GER number	Trade register number	211.415
entityIdentifierTradeRegisterNumber	Djibouti	Trade register number	Trade register number	952/B
entityIdentifierTradeRegisterNumber	Dominican Republic	Trade register number	Trade register number	33682 (Santo Domingo)
entityIdentifierRuc	Ecuador	RUC	VAT/Tax number	1768153530001
entityIdentifierNit	El Salvador	NIT	VAT/Tax number	06142102971036
entityIdentifierTradeRegisterNumber	Estonia	Trade register number	Trade register number	10005211
	Estonia	VAT number	VAT/Tax number	EE100254507
entityIdentifierEuropeanVatNumber	European Union	European VAT number	European VAT number	DE115235681
entityIdentifierBusinessId	Finland	Business ID	Company ID number	0112038-9
entityIdentifierSirenNumber	France	SIREN number	Company ID number	542051180
entityIdentifierSiretNumber	France	SIRET number	Company ID number	542051180-00066
entityIdentifierTradeRegisterNumber	Georgia	Trade register number	Trade register number	204378869
entityIdentifierTaxNumber	Germany	Tax number	VAT/Tax number	143/102/60208
entityIdentifierTradeRegisterNumber	Germany	Trade register number	Trade register number	HRB 100484 (Braunschweig)
entityIdentifierUstIdnr	Germany	USt-IdNr.	VAT/Tax number	DE 115235681
entityIdentifierRgdNumber	Ghana	RGD number	Trade register number	CS352862014
entityIdentifierTradeRegisterNumber	Gibraltar	Trade register number	Trade register number	85455
entityIdentifierAfm	Greece	AFM	VAT/Tax number	094049864
entityIdentifierGemiNumber	Greece	GEMI number	Trade register number	000296601000
entityIdentifierNit	Guatemala	NIT	VAT/Tax number	151030A141115
entityIdentifierPatente	Guatemala	Patente	Trade register number	36741371126
entityIdentifierTradeRegisterNumber	Guernsey	Trade register number	Trade register number	GG19528
entityIdentifierRccmNumber	Guinea	RCCM number	Trade register number	RCCM/GCKAL/02794A/2004
entityIdentifierTradeRegisterNumber	Guyana	Trade register number	Trade register number	573R
	Hong Kong	BRN number	Trade register number	00015833

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierTradeRegisterNumber	Hong Kong	Trade register number	Trade register number	F0010694
entityIdentifierStatisticalNumber	Hungary	Statistical number	Statistical number	23506181
entityIdentifierTradeRegisterNumber	Hungary	Trade register number	Trade register number	01-17-000705
	Hungary	VAT number	VAT/Tax number	10760798-2-44
entityIdentifierKennitala	Iceland	Kennitala	Trade register number	5505003530
	Iceland	VAT number	VAT/Tax number	08758
entityIdentifierCin	India	CIN	Trade register number	U16009WB1986PTC040023
entityIdentifierGstin	India	GSTIN	VAT/Tax number	27AAACT4060A1ZV
entityIdentifierLlpIn	India	LLPIN	Trade register number	AAA-6169
entityIdentifierPan	India	PAN	VAT/Tax number	AABCP5057C
	India	TIN	VAT/Tax number	27AAACI1681G1ZP
entityIdentifierTradeRegisterNumber	India	Trade register number	Trade register number	04-18863
	Indonesia	TIN	VAT/Tax number	01.001.629.3-051.000
entityIdentifierTradeRegisterNumber	Ireland	Trade register number	Trade register number	905143
entityIdentifierTradeRegisterNumber	Isle of Man	Trade register number	Trade register number	092157C
entityIdentifierTradeRegisterNumber	Israel	Trade register number	Trade register number	520013954
entityIdentifierCciaaNumber	Italy	CCIAA number	Trade register number	TO0091712
entityIdentifierCodiceFiscale	Italy	Codice fiscale	VAT/Tax number	00470400011
entityIdentifierCodiceIstat	Italy	Codice ISTAT	Statistical number	015021
entityIdentifierPartitaIva	Italy	Partita IVA	VAT/Tax number	00470400011
entityIdentifierBusinessRegistrationNumber	Jamaica	Business Registration Number	Trade register number	8139L
entityIdentifierCompanyNumber	Jamaica	Company number	Trade register number	63077L
entityIdentifierCorporateNumber	Japan	Corporate Number	VAT/Tax number	1180301018771
entityIdentifierTradeRegisterNumber	Jersey	Trade register number	Trade register number	107710
entityIdentifierBin	Kazakhstan	BIN	Company ID number	081140000436
entityIdentifierOkpo	Kazakhstan	OKPO	Statistical number	50534408

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierTaxNumber	Kenya	Tax number	VAT/Tax number	P051129820X
entityIdentifierCorporateRegisteredNumber	Korea (the Republic of)	Corporate registered number	Trade register number	1301110006246
	Korea (the Republic of)	VAT number	VAT/Tax number	124-81-00998
entityIdentifierTradeRegisterNumber	Kosovo	Trade register number	Trade register number	71102130
entityIdentifierTaxNumber	Kosovo	Tax number	VAT/Tax number	600052386
entityIdentifierUin	Kosovo	UIN	Trade register number	810413584
entityIdentifierNin	Kyrgyzstan	NIN	Trade register number	22717514
entityIdentifierTaxNumber	Kyrgyzstan	Tax number	VAT/Tax number	01505200110063
entityIdentifierTradeRegisterNumber	Latvia	Trade register number	Trade register number	43603021620
	Latvia	VAT number	VAT/Tax number	LV43603021620
	Lesotho	TIN	VAT/Tax number	200048452-3
entityIdentifierTradeRegisterNumber	Lesotho	Trade register number	Trade register number	1994165R
entityIdentifierTradeRegisterNumber	Liechtenstein	Trade register number	Trade register number	FL-0001.011.557-0
entityIdentifierTradeRegisterNumber	Lithuania	Trade register number	Trade register number	166451720
	Lithuania	VAT number	VAT/Tax number	LT664517219
entityIdentifierTaxNumber	Luxembourg	Tax number	VAT/Tax number	2001/2214/155
entityIdentifierTradeRegisterNumber	Luxembourg	Trade register number	Trade register number	B 160.772 (Luxembourg)
	Luxembourg	VAT number	VAT/Tax number	LU18928066
entityIdentifierNif	Madagascar	NIF	VAT/Tax number	3000003944
entityIdentifierRcsNumber	Madagascar	RCS number	Trade register number	2004B00553
entityIdentifierTradeRegisterNumber	Malaysia	Trade register number	Trade register number	20076-K
entityIdentifierTradeRegisterNumber	Mali	Trade register number	Trade register number	D0045C1
entityIdentifierTradeRegisterNumber	Malta	Trade register number	Trade register number	C 43070
	Malta	VAT number	VAT/Tax number	13240110
	Mauritius	BRN	Trade register number	C06000706
entityIdentifierTan	Mauritius	TAN	VAT/Tax number	20036902

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierTradeRegisterNumber	Mauritius	Trade register number	Trade register number	C13145
entityIdentifierRfc	Mexico	RFC	VAT/Tax number	PME 380607 P35
entityIdentifierChamberOfCommerceNumber	Middle East	Chamber of commerce number	Trade register number	13996
entityIdentifierCommercialRegisterNumber	Middle East	Commercial register number	Trade register number	2052000234
entityIdentifierIndustrialLicense	Middle East	Industrial license	Trade register number	93860
	Middle East	VAT number	VAT/Tax number	300050206400003
entityIdentifierIdno	Moldova	IDNO	Trade register number	1003600005148
entityIdentifierRciNumber	Monaco	RCI number	Trade register number	56S00448
entityIdentifierRegistarskiBroj	Montenegro	Registarski broj	Trade register number	02002230
entityIdentifierIce	Morocco	ICE	Company ID number	1526005000070
entityIdentifierTradeRegisterNumber	Morocco	Trade register number	Trade register number	91-83 (MOHAMEDIA)
entityIdentifierTradeRegisterNumber	Myanmar	Trade register number	Trade register number	445/2004-2005
entityIdentifierFin	Namibia	FIN	VAT/Tax number	4260139888
entityIdentifierTradeRegisterNumber	Namibia	Trade register number	Trade register number	CY/2006/0062
entityIdentifierTradeRegisterNumber	Nepal	Trade register number	Trade register number	928D1R
entityIdentifierCocNumber	Netherlands	CoC-number	Trade register number	34179503
entityIdentifierLocationNumber	Netherlands	Location number	Company ID number	000020944705
entityIdentifierRsin	Netherlands	RSIN	VAT/Tax number	811786523
entityIdentifierCompanyNumber	New Zealand	Company number	Trade register number	1166320
entityIdentifierNzbn	New Zealand	NZBN	Company ID number	9429036748471
entityIdentifierNif	Niger	NIF	VAT/Tax number	10000
entityIdentifierRccmNumber	Niger	RCCM number	Trade register number	A/82/2005
entityIdentifierStatisticalNumber	Niger	Statistical number	Company ID number	905
entityIdentifierRcNumber	Nigeria	RC number	Trade register number	208767R
entityIdentifierPin	Republic of North Macedonia	PIN	Statistical number	6246648
	Republic of North Macedonia	TIN	VAT/Tax number	MK4030007641185

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierOrganizationNumber	Norway	Organization number	Company ID number	923 609 016
entityIdentifierCrNumber	Oman	CR number	Trade register number	1703412
entityIdentifierTradeRegisterNumber	Pakistan	Trade register number	Trade register number	0004460
entityIdentifierTradeRegisterNumber	Panama	Trade register number	Trade register number	155584709
entityIdentifierTradeRegisterNumber	Panama	Trade register number	Trade register number	466377RPP
entityIdentifierTradeRegisterNumber	Papua New Guinea	Trade register number	Trade register number	1-125
entityIdentifierRuc	Paraguay	RUC	VAT/Tax number	80009310-0
entityIdentifierRuc	Peru	RUC	VAT/Tax number	20259829594
entityIdentifierSecRegistrationNumber	Philippines	SEC registration number	Trade register number	PW00000277
	Philippines	TRN	Trade register number	00140569
entityIdentifierNcrNumber	Poland	NCR number	Trade register number	0000028860
entityIdentifierNip	Poland	NIP	VAT/Tax number	7740001454
entityIdentifierRegonNumber	Poland	REGON number	Statistical number	610188201
entityIdentifierNif	Portugal	NIF	VAT/Tax number	500715505
entityIdentifierTradeRegisterNumber	Qatar	Trade register number	Trade register number	21
entityIdentifierTradeRegisterNumber	Romania	Trade register number	Trade register number	J40/8302/1997
	Romania	VAT number	VAT/Tax number	1590082
entityIdentifierInn	Russian Federation	INN	VAT/Tax number	7703585780
entityIdentifierOgrn	Russian Federation	OGRN	Trade register number	1067746393780
entityIdentifierOgrnip	Russian Federation	OGRNIP	Trade register number	304264329900257
entityIdentifierOkpo	Russian Federation	OKPO	Statistical number	94102445
entityIdentifierEnterpriseCode	Rwanda	Enterprise code	Company ID number	100004348
entityIdentifierNinea	Senegal	NINEA	Company ID number	000049861
entityIdentifierRccmNumber	Senegal	RCCM number	Trade register number	SN.DKR.2014.B.14908
entityIdentifierTradeRegisterNumber	Serbia	Trade register number	Trade register number	08049904
	Serbia	VAT number	VAT/Tax number	104052135

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierBusinessRegistrationNumber	Seychelles	Business Registration Number	Trade register number	C841349
	Seychelles	TIN	VAT/Tax number	151141300
entityIdentifierUen	Singapore	UEN	Trade register number	201017488D
entityIdentifierTradeRegisterNumber	Slovakia	Trade register number	Trade register number	00168912
	Slovakia	VAT number	VAT/Tax number	2020382111
entityIdentifierTradeRegisterNumber	Slovenia	Trade register number	Trade register number	5025796000
	Slovenia	VAT number	VAT/Tax number	80267432
entityIdentifierEnterpriseNumber	South Africa	Enterprise number	Trade register number	2002/042809/23
entityIdentifierTaxNumber	South Africa	Tax number	VAT/Tax number	9483281144
	South Africa	VAT number	VAT/Tax number	4900202419
entityIdentifierNif	Spain	NIF	VAT/Tax number	A28015865
entityIdentifierTradeRegisterNumber	Sri Lanka	Trade register number	Trade register number	PB67
entityIdentifierCfarNumber	Sweden	CFAR-number	Statistical number	46238176
entityIdentifierTradeRegisterNumber	Sweden	Trade register number	Trade register number	556012-5790
entityIdentifierChNumber	Switzerland	CH-number	Trade register number	CH-550.0.067.293-5
entityIdentifierUid	Switzerland	UID	Company ID number	CHE-105.909.036
entityIdentifierTradeRegisterNumber	Taiwan (Province of China)	Trade register number	Trade register number	04541302
entityIdentifierEin	Tajikistan	EIN	Trade register number	0110000019
entityIdentifierInn	Tajikistan	INN	VAT/Tax number	010017585
	Tanzania, United Republic of	TIN	VAT/Tax number	100093697
entityIdentifierTradeRegisterNumber	Tanzania, United Republic of	Trade register number	Trade register number	2497
entityIdentifierTradeRegisterNumber	Thailand	Trade register number	Trade register number	0107544000108
entityIdentifierTradeRegisterNumber	Togo	Trade register number	Trade register number	1986-B-1575
entityIdentifierTradeRegisterNumber	Tunisia	Trade register number	Trade register number	B169701997
entityIdentifierMersisNumber	Turkey	Mersis number	Company ID number	0570002057500012

Key (Identifier ID)	Country or Region	Identifier Name	Type	Example
entityIdentifierTradeRegisterNumber	Turkey	Trade register number	Trade register number	85714 (Istanbul Merkez)
entityIdentifierTradeRegisterNumber	Uganda	Trade register number	Trade register number	80010002665812
entityIdentifierEdrpouCode	Ukraine	EDRPOU code	Statistical number	20077720
	Ukraine	VAT number	VAT/Tax number	200777226658
entityIdentifierLicenseNumber	United Arab Emirates	License number	Trade register number	230704
sedol	United Kingdom of Great Britain and Northern Ireland	Stock exchange daily official list (SEDOL)	Securities number	0540528
entityIdentifierCharityNumber	United Kingdom of Great Britain and Northern Ireland	Charity number	Trade register number	209131
entityIdentifierCompanyNumber	United Kingdom of Great Britain and Northern Ireland	Company number	Trade register number	04366849
	United Kingdom of Great Britain and Northern Ireland	VAT number	VAT/Tax number	GB629280722
cusip	United States of America	CUSIP number	Securities number	345370
	United States of America	Cortera LINK ID	IP identification number	138303076
entityIdentifierEin	United States of America	EIN	VAT/Tax number	71-0415188
entityIdentifierNaicNumber	United States of America	NAIC number	Company ID number	25178
entityIdentifierTradeRegisterNumber	United States of America (Puerto Rico)	Trade register number	Trade register number	8-611
entityIdentifierRuc	Uruguay	RUC	VAT/Tax number	210592210014
entityIdentifierRut	Uruguay	RUT	VAT/Tax number	215051740010
entityIdentifierTradeRegisterNumber	Uzbekistan	Trade register number	Trade register number	200941525
entityIdentifierRif	Venezuela	RIF	VAT/Tax number	J-00095036-9
entityIdentifierRnc	Venezuela	RNC	Trade register number	1139727001241350000
entityIdentifierTradeRegisterNumber	Vietnam	Trade register number	Trade register number	0100100079
entityIdentifierTpin	Zambia	TPIN	VAT/Tax number	1001958276
entityIdentifierTradeRegisterNumber	Zambia	Trade register number	Trade register number	120090078633

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